Convergence of Stochastic Gradient Descent with Momentum

An expected-decrease analysis in the style of SGD, with complete proofs, and a map of when convergence holds

SGDM_Proofs project

October 2026

Abstract

Stochastic gradient descent with momentum (SGDM, also called the stochastic heavy-ball method) is a default optimizer in machine learning, yet its convergence proof is far less familiar than the textbook proof for plain SGD. The obstacle is that momentum lets the iterates overshoot, so the function value $f(x_k)$ need not decrease—not even in expectation, and not even without noise. This report shows how to recover an SGD-style *expected decrease inequality* for SGDM by measuring progress with a potential that combines the function value at an extrapolated point with a kinetic-energy term. From this single inequality we derive, with every step written out: (i) convergence to approximate stationary points of smooth nonconvex functions at the same $O(1/\sqrt{K})$ rate as SGD; (ii) convergence with diminishing step sizes; (iii) linear convergence up to a noise floor, and an $O(1/k)$ rate with decaying steps, under the Polyak–Łojasiewicz condition; and (iv) an $O(1/\sqrt{K})$ rate for convex functions under a milder step-size condition. We then map out *when* these guarantees hold: we compute the noise floor exactly on quadratics, derive the exact stability region of the heavy-ball method on quadratics, prove that without noise a mechanical energy decreases for every smooth function when $\alpha < 2(1-\beta)/L$, and verify by hand a classical counterexample in which momentum parameters tuned for quadratics make the method cycle on a strongly convex function. Numerical experiments illustrate each statement. Only multivariable calculus, linear algebra and basic probability are assumed.

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1 Introduction

1.1 The problem and the two algorithms

We want to solve the unconstrained minimization problem

$$\min_{x\in\mathbb{R}^d} f(x), \tag{1.1}$$

where $f:\mathbb{R}^d\to\mathbb{R}$ is differentiable, but its gradient ∇f is not available exactly. Instead, at any point x we can ask for a *stochastic gradient* $g(x;\xi)$: a random vector, depending on a random sample ξ, which is correct *on average*, $\mathbb{E}_\xi[g(x;\xi)]=\nabla f(x)$. This is the situation in machine learning, where $f(x)=\frac{1}{n}\sum_{i=1}^n f_i(x)$ is the average loss over n training examples and $g(x;\xi)=\nabla f_\xi(x)$ is the gradient of the loss on one randomly chosen example (or on a random mini-batch).

SGD. Stochastic gradient descent starts at a point x_0 and repeats

$$x_{k+1}=x_k-\alpha\, g_k,\qquad g_k:=g(x_k;\xi_k),\qquad k=0,1,2,\dots \tag{1.2}$$

where $\alpha>0$ is the *step size* (learning rate) and $\xi_0,\xi_1,\dots$ are independent, identically distributed samples.

SGDM. SGD with momentum—also called the *stochastic heavy-ball method*, after Polyak's heavy-ball method [11]—adds a fraction $\beta\in[0,1)$ of the previous displacement:

$$x_{k+1}=x_k-\alpha\, g_k+\beta\,(x_k-x_{k-1}),\qquad x_{-1}:=x_0. \tag{1.3}$$

Physically, x_k is the position of a heavy ball rolling down the graph of f: the term $\beta(x_k-x_{k-1})$ is its inertia (it keeps moving in the direction it was already moving), and $1-\beta$ plays the role of friction. With $\beta=0$ we recover SGD. In deep learning, (1.3)—usually implemented in an equivalent "momentum buffer" form (Section 4.1)—with $\beta=0.9$ is a standard default [15].

1.2 What "convergence" means here

For a nonconvex f there is no efficient method that is guaranteed to find a global minimizer, so the standard goal is an approximate *stationary point*, a point with a small gradient. Since the iterates are random, guarantees are stated in expectation. A typical statement has the form

$$\min_{0\le k<K}\mathbb{E}\|\nabla f(x_k)\|^2\ \le\ (\text{a quantity that tends to 0 as } K\to\infty),$$

and the speed at which the right-hand side decreases in the number of iterations K is the *rate*. For special classes of functions (Polyak–Łojasiewicz or convex) we will also bound the optimality gap $\mathbb{E}[f(x_k)]-f^\star$, where $f^\star=\inf f$ (for convex functions, at the averaged iterate).

1.3 The difficulty, and the idea in one line

For SGD on smooth, possibly nonconvex functions, the textbook proof [2, 5] rests on one inequality, the *expected decrease*:

$$\mathbb{E}\big[f(x_{k+1})\,\big|\,\text{history}\big]\ \le\ f(x_k)-\frac{\alpha}{2}\|\nabla f(x_k)\|^2+\frac{L\alpha^2}{2}\sigma^2\qquad(\alpha\le 1/L), \tag{1.4}$$

where L is the smoothness constant of f and σ^2 bounds the variance of the stochastic gradients (Section 3). In words: on average, one step decreases f by an amount proportional to the squared gradient, minus a price for the noise that is *quadratic* in the step size. Everything else—rates, step-size rules—follows by summing (1.4) over k.

For SGDM, inequality (1.4) is *false*, even without noise: momentum can carry the iterate uphill (in Example 4.2 the method sits exactly at the minimizer and is pushed away by its own inertia). It can be guaranteed at a given step when $\alpha \le \frac{1}{2L}$ and the momentum points sufficiently downhill, $\langle \nabla f(x_k),\, x_k - x_{k-1}\rangle \le -L\beta\|x_k - x_{k-1}\|^2$ (Proposition 4.3); after an overshoot no such guarantee is available, and the algorithm cannot control when overshoots happen. The function value is the wrong yardstick. The idea of this report is to measure progress with a *potential*

$$\Phi_k = f(z_k) - f^\star + \frac{L\beta^2}{2(1-\beta)^2}\|x_k - x_{k-1}\|^2, \qquad z_k = x_k + \frac{\beta}{1-\beta}\,(x_k - x_{k-1}),$$

where z_k is an extrapolated point—the place where the ball would come to rest if the gradient force were switched off and only friction acted (Section 4.4)—and the second term is a kinetic energy. We prove (Theorem 5.3) that Φ_k obeys the exact analogue of (1.4):

$$\mathbb{E}\big[\Phi_{k+1} \,\big|\, \text{history}\big] \;\le\; \Phi_k - \frac{\gamma}{4}\|\nabla f(x_k)\|^2 + L\gamma^2\sigma^2, \qquad \gamma := \frac{\alpha}{1-\beta}, \tag{1.5}$$

$$\text{whenever the step size satisfies } \alpha \le \frac{(1-\beta)^2}{2L}.$$

Once (1.5) is available, all the standard SGD conclusions transfer to SGDM by the same telescoping arguments, with the *effective step size* $\gamma = \alpha/(1-\beta)$ playing the role of α.

1.4 Summary of results

Table 1 lists what we prove. Throughout, $\Delta := f(x_0) - f^\star$, $\gamma = \alpha/(1-\beta)$, and "nonconvex" means only that f is L-smooth and bounded below. In the table, μ is the constant of the Polyak–Łojasiewicz (PL) inequality (Definition 2.6), $x^\star$ is a minimizer, and $\bar{x}_K = \frac{1}{K}\sum_{k<K} x_k$.

1.5 How to read this report

Prerequisites. Multivariable calculus (gradients, the chain rule, the fundamental theorem of calculus), linear algebra (inner products, norms, eigenvalues of 2×2 matrices) and probability (expectation, independence, and conditional expectation at the level of "average over what is not yet known"). Section 2 collects every tool we use, with proofs; later proofs cite these facts explicitly.

Road map. Section 3 reviews the SGD analysis, which serves as the template. Section 4 introduces SGDM, explains why the function value is the wrong yardstick, and introduces the extrapolated point z_k. Section 5, the heart of the report, proves the expected decrease inequality (1.5). Section 6 turns it into convergence theorems. Section 7 answers the question "under what circumstances does SGDM converge?", including exact computations that show which parts of the conditions cannot be removed. Section 8 illustrates everything numerically; the code that produces every figure is in the `experiments/` folder of the repository.

Setting	Step-size condition	Guarantee	Where
Nonconvex	$\alpha \le \frac{(1-\beta)^2}{2L}$	$\frac{1}{K}\sum_{k<K}\mathbb{E}\|\nabla f(x_k)\|^2 \le \frac{4\Delta}{\gamma K} + 4L\gamma\sigma^2$	Theorem 6.1
Nonconvex, tuned γ	$\gamma = \min\{\frac{1-\beta}{2L}, \frac{1}{\sigma}\sqrt{\frac{\Delta}{LK}}\}$	$\frac{1}{K}\sum_{k<K}\mathbb{E}\|\nabla f(x_k)\|^2 \le \frac{8L\Delta}{(1-\beta)K} + 8\sigma\sqrt{\frac{L\Delta}{K}}$	Corollary 6.2
Nonconvex, decaying	$\gamma_k \le \frac{1-\beta}{2L}$, $\sum\gamma_k = \infty$, $\sum\gamma_k^2 < \infty$	$\liminf_k \|\nabla f(x_k)\| = 0$ almost surely	Theorem 6.4
No noise ($\sigma = 0$)	$\alpha = \frac{(1-\beta)^2}{2L}$	Φ_k non-increasing; $\min_{k<K}\|\nabla f(x_k)\|^2 \le \frac{8L\Delta}{(1-\beta)K}$	Corollary 6.3
No noise, smooth	$\alpha < \frac{2(1-\beta)}{L}$	energy decreases; $\nabla f(x_k) \to 0$	Proposition 7.3
PL inequality	$\alpha \le \frac{(1-\beta)^2}{2L}$	$\mathbb{E}f(x_k) - f^\star \le 2\big(1-\frac{\gamma\mu}{4}\big)^k\Delta + \frac{8L\gamma\sigma^2}{\mu}$	Theorem 6.6
PL, decaying	$\gamma_k = \frac{8}{\mu(k+a)}$, $a \ge \frac{16L}{\mu(1-\beta)}$	$\mathbb{E}f(x_k) - f^\star = O(1/k)$	Corollary 6.7
Convex	$\alpha \le \frac{1-\beta}{2L}$	$\mathbb{E}f(\bar{x}_K) - f^\star \le \frac{\|x_0-x^\star\|^2}{\gamma K} + \frac{2\beta\Delta}{(1-\beta)K} + \gamma\sigma^2$	Theorem 6.8

Table 1: Summary of the guarantees proved in this report (all for SGDM in heavy-ball form with momentum $\beta \in [0, 1)$; for $\beta = 0$ they reduce to the corresponding SGD results up to constants). The question "under what circumstances?" is discussed in Section 7: there we show that the noise floor $\propto \gamma\sigma^2$ is unavoidable with a constant step (Proposition 7.1), that too large a step size makes the method diverge or cycle (Proposition 7.2, Example 7.5), and how the admissible step sizes shrink as $\beta \to 1$ (Figure 3).

Relation to the literature. The extrapolated-point ("virtual sequence") technique and potentials of this type are classical in analyses of momentum methods [4, 16, 17, 9, 14]. The aim here is a self-contained, fully explicit version—one potential, one key inequality, and explicit constants—written so that each line can be checked by hand. Constants are chosen for readability, not optimized.

2 Preliminaries

This section collects every tool used later. Each fact comes with a short proof, so that the main arguments can be checked line by line.

2.1 Notation

Vectors live in $\mathbb{R}^d$. For $a, b \in \mathbb{R}^d$, $\langle a, b\rangle = \sum_{i=1}^d a_i b_i$ is the inner product and $\|a\| = \sqrt{\langle a, a\rangle}$ the Euclidean norm. For $f : \mathbb{R}^d \to \mathbb{R}$ we write $f^\star := \inf_{x\in\mathbb{R}^d} f(x)$ (possibly not attained). Sums $\sum_{k<K}$ run over $k = 0, 1, \ldots, K-1$. For SGDM we will use the following names; each is defined precisely where it first appears in the text:

$\alpha > 0$, $\beta \in [0,1)$	step size and momentum parameter
$\gamma = \alpha/(1-\beta)$	effective step size
$v_k = x_k - x_{k-1}$	velocity (last displacement), $v_0 = 0$
$z_k = x_k + \frac{\beta}{1-\beta} v_k$	extrapolated point
$c_\beta = \frac{L\beta^2}{2(1-\beta)^2}$	weight of the kinetic energy
$\Phi_k = f(z_k) - f^\star + c_\beta \|v_k\|^2$	potential
$\varepsilon_k = g_k - \nabla f(x_k)$	gradient noise
$\Delta = f(x_0) - f^\star$	initial optimality gap

2.2 Elementary inequalities

Lemma 2.1 (Elementary facts)**.** *For all $a, b \in \mathbb{R}^d$:*

(a) (Cauchy–Schwarz) $|\langle a,\, b\rangle| \le \|a\|\|b\|$.

(b) (Young) *For every $\rho > 0$:* $\langle a,\, b\rangle \le \frac{\rho}{2}\|a\|^2 + \frac{1}{2\rho}\|b\|^2$.

(c) (Polarization) $-\langle a,\, b\rangle = \frac{1}{2}\|a-b\|^2 - \frac{1}{2}\|a\|^2 - \frac{1}{2}\|b\|^2$.

(d) (Convexity of $\|\cdot\|^2$) *For every $\theta \in [0,1]$:* $\|\theta a + (1-\theta)b\|^2 \le \theta\|a\|^2 + (1-\theta)\|b\|^2$.

(e) $\|a+b\|^2 \le 2\|a\|^2 + 2\|b\|^2$.

(f) (Geometric series) *For $q \in [0,1)$ and $k \ge 1$:* $\sum_{j=0}^{k-1} q^j = \frac{1-q^k}{1-q} \le \frac{1}{1-q}$.

Proof. (a) is the Cauchy–Schwarz inequality from linear algebra. (b) $0 \le \left\|\sqrt{\rho}\, a - b/\sqrt{\rho}\right\|^2 = \rho\|a\|^2 - 2\langle a,\, b\rangle + \|b\|^2/\rho$; rearrange and divide by 2. (c) Expand $\|a-b\|^2 = \|a\|^2 - 2\langle a,\, b\rangle + \|b\|^2$ and rearrange. (d) Expanding both sides,

$$\theta\|a\|^2 + (1-\theta)\|b\|^2 - \|\theta a + (1-\theta)b\|^2 = \theta(1-\theta)\big(\|a\|^2 - 2\langle a,\, b\rangle + \|b\|^2\big) = \theta(1-\theta)\|a-b\|^2 \ge 0.$$

(e) Apply (d) with $\theta = \frac{1}{2}$ to the vectors $2a$ and $2b$: $\|a+b\|^2 = \|\frac{1}{2}(2a) + \frac{1}{2}(2b)\|^2 \le \frac{1}{2}\|2a\|^2 + \frac{1}{2}\|2b\|^2$. (f) Multiply $\sum_{j=0}^{k-1} q^j$ by $1-q$; the sum telescopes to $1 - q^k \le 1$. □

2.3 Smooth functions and the descent lemma

Definition 2.2 (L-smoothness)**.** A differentiable $f : \mathbb{R}^d \to \mathbb{R}$ is *L-smooth* ($L > 0$) if its gradient is L-Lipschitz: $\|\nabla f(x) - \nabla f(y)\| \le L\|x - y\|$ for all $x, y \in \mathbb{R}^d$.

For twice differentiable f, L-smoothness is equivalent to all eigenvalues of the Hessian $\nabla^2 f(x)$ lying in $[-L, L]$; for example $f(x) = \frac{h}{2}x^2$ is $|h|$-smooth. The single most important consequence is that an L-smooth function lies below a parabola around every point.

Lemma 2.3 (Descent lemma)**.** *If f is L-smooth, then for all $x, y \in \mathbb{R}^d$,*

$$f(y) \le f(x) + \langle \nabla f(x),\, y - x\rangle + \frac{L}{2}\|y - x\|^2. \tag{2.1}$$

Proof. Let $\varphi(t) = f(x + t(y-x))$ for $t \in [0,1]$. By the chain rule $\varphi'(t) = \langle \nabla f(x + t(y-x)),\, y - x\rangle$, which is continuous because ∇f is Lipschitz. By the fundamental theorem of calculus, $f(y) - f(x) =$

$\varphi(1) - \varphi(0) = \int_0^1 \varphi'(t)\,dt$. Adding and subtracting $\langle \nabla f(x),\, y - x\rangle$ inside the integral,

$$\begin{aligned} f(y) - f(x) &= \langle \nabla f(x),\, y - x\rangle + \int_0^1 \langle \nabla f(x + t(y-x)) - \nabla f(x),\, y - x\rangle\, dt \\ &\le \langle \nabla f(x),\, y - x\rangle + \int_0^1 \|\nabla f(x + t(y-x)) - \nabla f(x)\|\, \|y - x\|\, dt \\ &\le \langle \nabla f(x),\, y - x\rangle + \int_0^1 L\, t \|y - x\|^2\, dt = \langle \nabla f(x),\, y - x\rangle + \frac{L}{2}\|y - x\|^2, \end{aligned}$$

where we used Cauchy–Schwarz (Lemma 2.1(a)) and then L-smoothness. □

Lemma 2.4 (Gradients are controlled by the optimality gap)**.** *If f is L-smooth and $f^\star > -\infty$, then $\|\nabla f(x)\|^2 \le 2L(f(x) - f^\star)$ for every $x \in \mathbb{R}^d$.*

Proof. Apply Lemma 2.3 with $y = x - \frac{1}{L}\nabla f(x)$: $f^\star \le f(y) \le f(x) - \frac{1}{L}\|\nabla f(x)\|^2 + \frac{1}{2L}\|\nabla f(x)\|^2 = f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$. Rearrange. □

2.4 Convexity and the Polyak–Łojasiewicz inequality

Definition 2.5. A differentiable f is *convex* if $f(y) \ge f(x) + \langle \nabla f(x),\, y - x\rangle$ for all x, y, and *μ-strongly convex* ($\mu > 0$) if $f(y) \ge f(x) + \langle \nabla f(x),\, y - x\rangle + \frac{\mu}{2}\|y - x\|^2$ for all x, y.

(For differentiable functions these first-order inequalities are equivalent to the usual definitions of convexity; we take them as definitions.)

Definition 2.6 (PL inequality)**.** An f with $f^\star > -\infty$ satisfies the *Polyak–Łojasiewicz (PL) inequality* with constant $\mu > 0$ if

$$\|\nabla f(x)\|^2 \ge 2\mu(f(x) - f^\star) \qquad \text{for all } x \in \mathbb{R}^d. \tag{2.2}$$

The PL inequality says that the gradient can be small only where the function value is nearly optimal; in particular every stationary point is a global minimizer. It does *not* require convexity.

Lemma 2.7. (a) *A μ-strongly convex function satisfies the PL inequality with constant μ.*

(b) *If f is L-smooth and satisfies PL with constant μ, and ∇f is not identically zero, then $\mu \le L$.*

(c) *The function $f(x) = x^2 + 3\sin^2 x$ on $\mathbb{R}$ is nonconvex, 8-smooth, and satisfies PL (with $\mu = 1/32$, see [6]); so does $\sum_{i=1}^d (x_i^2 + 3\sin^2 x_i)$ on $\mathbb{R}^d$.*

Proof. (a) Fix x. The right-hand side of the strong-convexity inequality is a quadratic function of y, minimized at $y = x - \frac{1}{\mu}\nabla f(x)$ with minimum value $f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$. Hence $f(y) \ge f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$ for every y; taking the infimum over y gives $f^\star \ge f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$, which is (2.2).

(b) Pick x with $\nabla f(x) \ne 0$. By Lemma 2.4 and (2.2), $\frac{1}{2L}\|\nabla f(x)\|^2 \le f(x) - f^\star \le \frac{1}{2\mu}\|\nabla f(x)\|^2$; divide by $\|\nabla f(x)\|^2 > 0$.

(c) $f''(x) = 2 + 6\cos 2x \in [-4, 8]$, so f is 8-smooth and (as f'' takes negative values) nonconvex. The PL constant is computed in [6]; for a sum of functions of separate coordinates, (2.2) holds coordinate-wise and therefore for the sum. □

Convention. Whenever we assume the PL inequality, we also assume $\mu \le L$. This is no restriction: if $\nabla f \not\equiv 0$ it holds automatically by (b), and in general the PL inequality with constant μ implies the PL inequality with every smaller constant, in particular with $\min\{\mu, L\}$.

2.5 Randomness and conditional expectation

The probabilistic model. The samples $\xi_0, \xi_1, \xi_2, \dots$ are independent and identically distributed (i.i.d.), each distributed like a generic sample ξ, and the starting point x_0 is deterministic. The stochastic gradient oracle $g(x;\xi)$ satisfies, for every *fixed* $x \in \mathbb{R}^d$,

$$\mathbb{E}_\xi[g(x;\xi)] = \nabla f(x), \qquad \mathbb{E}_\xi\|g(x;\xi) - \nabla f(x)\|^2 \le \sigma^2. \tag{2.3}$$

What is known at time k. Every quantity computed by the algorithm *before* it draws ξ_k—the points $x_1, \dots, x_k$, and hence $\nabla f(x_k)$, $f(x_k)$, $v_k = x_k - x_{k-1}$, z_k, Φ_k—is a function of $\xi_0, \dots, \xi_{k-1}$. We say such a quantity is *known at time k*. The quantities g_k, x_{k+1}, z_{k+1}, Φ_{k+1} are not known at time k (they depend on ξ_k). We write

$$\mathbb{E}_k[Y] := \mathbb{E}\big[Y \,\big|\, \xi_0, \dots, \xi_{k-1}\big]$$

for the conditional expectation of a random variable Y given the history up to time k. Intuitively, $\mathbb{E}_k[Y]$ averages Y over the samples that have not been drawn yet, holding the history fixed; it is itself a random quantity (a function of the history). We use only the following standard rules (valid whenever the expectations involved are finite):

(R1) *Linearity:* $\mathbb{E}_k[aY + bZ] = a\mathbb{E}_k[Y] + b\mathbb{E}_k[Z]$ for constants a, b.

(R2) *Known factors come out:* if U is known at time k then $\mathbb{E}_k[U\,Y] = U\,\mathbb{E}_k[Y]$ and $\mathbb{E}_k[U] = U$; for vectors, $\mathbb{E}_k\langle U, Y\rangle = \langle U, \mathbb{E}_k[Y]\rangle$.

(R3) *Tower property:* $\mathbb{E}\big[\mathbb{E}_k[Y]\big] = \mathbb{E}[Y]$.

(R4) *Monotonicity:* if $Y \le Z$ then $\mathbb{E}_k[Y] \le \mathbb{E}_k[Z]$.

(R5) *Freezing:* if U is known at time k, then $\mathbb{E}_k\big[h(U, \xi_k)\big] = H(U)$, where $H(u) := \mathbb{E}_\xi[h(u, \xi)]$ is the ordinary expectation computed with u held fixed. (This uses that ξ_k is independent of $\xi_0, \dots, \xi_{k-1}$.)

Applying (R5) to (2.3) with $U = x_k$ gives the two properties that the analysis actually uses:

$$\mathbb{E}_k[g_k] = \nabla f(x_k), \qquad \mathbb{E}_k\|\varepsilon_k\|^2 \le \sigma^2, \qquad \text{where } \varepsilon_k := g_k - \nabla f(x_k). \tag{2.4}$$

Lemma 2.8 (Variance decomposition)**.** *Let Y be a random vector with $\mathbb{E}_k\|Y\|^2 < \infty$ whose conditional mean $m := \mathbb{E}_k[Y]$ is known at time k (as it always is). Then $\mathbb{E}_k\|Y\|^2 = \|m\|^2 + \mathbb{E}_k\|Y - m\|^2$. In particular $\mathbb{E}_k\|g_k\|^2 = \|\nabla f(x_k)\|^2 + \mathbb{E}_k\|\varepsilon_k\|^2 \le \|\nabla f(x_k)\|^2 + \sigma^2$.*

Proof. $\|Y\|^2 = \|m + (Y-m)\|^2 = \|m\|^2 + 2\langle m, Y-m\rangle + \|Y-m\|^2$. Take $\mathbb{E}_k$: by (R2), $\mathbb{E}_k\langle m, Y-m\rangle = \langle m, \mathbb{E}_k[Y] - m\rangle = 0$. The second claim is the case $Y = g_k$, $m = \nabla f(x_k)$, combined with (2.4). □

Remark 2.9 (Mini-batches). If g_k is the average of B stochastic gradients computed with independent samples at the same point x_k, then (2.4) holds with σ^2 replaced by σ^2/B: the noises are independent with mean zero, so all cross terms in $\mathbb{E}_k\|\frac{1}{B}\sum_{b=1}^{B}\varepsilon_{k,b}\|^2$ vanish and B terms of size at most σ^2/B^2 remain.

2.6 Standing assumptions

Assumption A1 (Smoothness)**.** f is L-smooth (Definition 2.2).

Assumption A2 (Lower bound)**.** $f^\star = \inf_x f(x) > -\infty$.

Assumption A3 (Unbiased gradients)**.** $\mathbb{E}_k[g_k] = \nabla f(x_k)$ for every $k \geq 0$.

Assumption A4 (Bounded variance)**.** $\mathbb{E}_k\|g_k - \nabla f(x_k)\|^2 \leq \sigma^2$ for every $k \geq 0$.

Assumptions A3 and A4 hold under the model (2.3) by (2.4); the proofs below use only these conditional versions, so they also cover other sampling schemes for which Assumptions A3 and A4 hold. The case $\sigma = 0$ is deterministic (exact gradients).

Remark 2.10 (All expectations are finite). Under Assumptions A1 to A4, every quantity whose expectation we take is integrable. Indeed, by induction on k one has $\mathbb{E}\|x_k\|^2 < \infty$: if this holds up to k, then $\mathbb{E}\|g_k\|^2 = \mathbb{E}[\mathbb{E}_k\|g_k\|^2] \leq \mathbb{E}\|\nabla f(x_k)\|^2 + \sigma^2 \leq 2\|\nabla f(0)\|^2 + 2L^2\mathbb{E}\|x_k\|^2 + \sigma^2 < \infty$ (Lemma 2.8, (R3), L-smoothness and Lemma 2.1(e)), and x_{k+1} is a linear combination of x_k, x_{k-1} and g_k. Finally $f^\star \leq f(x) \leq f(0) + \|\nabla f(0)\|\|x\| + \frac{L}{2}\|x\|^2$ by Lemma 2.3 and Cauchy–Schwarz, so $\mathbb{E}|f(x_k)| < \infty$, and similarly for z_k and Φ_k. We will not mention integrability again.

3 Warm-up: the expected-decrease analysis of SGD

We first prove the standard results for SGD (1.2). The structure of these proofs is exactly what we will reproduce for SGDM, so it is worth isolating.

3.1 One step of SGD

Lemma 3.1 (Expected decrease for SGD)**.** *Under Assumptions A1, A3 and A4, the SGD iterates satisfy, for every $k \geq 0$,*

$$\mathbb{E}_k[f(x_{k+1})] \leq f(x_k) - \alpha\Big(1 - \frac{L\alpha}{2}\Big)\|\nabla f(x_k)\|^2 + \frac{L\alpha^2}{2}\sigma^2. \tag{3.1}$$

If $\alpha \leq 1/L$, then $\mathbb{E}_k[f(x_{k+1})] \leq f(x_k) - \frac{\alpha}{2}\|\nabla f(x_k)\|^2 + \frac{L\alpha^2}{2}\sigma^2$.

Proof. Apply the descent lemma (Lemma 2.3) with $x = x_k$ and $y = x_{k+1} = x_k - \alpha g_k$:

$$f(x_{k+1}) \leq f(x_k) - \alpha\langle\nabla f(x_k),\, g_k\rangle + \frac{L\alpha^2}{2}\|g_k\|^2.$$

Take $\mathbb{E}_k$ of both sides (R4). Since $f(x_k)$ and $\nabla f(x_k)$ are known at time k, rules (R1)–(R2) and Assumption A3 give $\mathbb{E}_k\langle\nabla f(x_k),\, g_k\rangle = \langle\nabla f(x_k),\, \mathbb{E}_k g_k\rangle = \|\nabla f(x_k)\|^2$, and Lemma 2.8 with Assumption A4 gives $\mathbb{E}_k\|g_k\|^2 \leq \|\nabla f(x_k)\|^2 + \sigma^2$. Hence

$$\mathbb{E}_k[f(x_{k+1})] \leq f(x_k) - \alpha\|\nabla f(x_k)\|^2 + \frac{L\alpha^2}{2}(\|\nabla f(x_k)\|^2 + \sigma^2),$$

which is (3.1). If $\alpha \leq 1/L$ then $1 - \frac{L\alpha}{2} \geq \frac{1}{2}$. □

Reading the inequality

The term $-\frac{\alpha}{2}\|\nabla f(x_k)\|^2$ is the *progress* (first order in α); the term $\frac{L\alpha^2}{2}\sigma^2$ is the *price of noise* (second order in α). One step decreases f in expectation whenever $\|\nabla f(x_k)\|^2 > L\alpha\sigma^2$, i.e. outside a "noise region" whose size shrinks with the step size. Near a stationary point the gradient is small, the noise wins, and with a constant step the iterates keep fluctuating.

3.2 Nonconvex functions

Theorem 3.2 (SGD, nonconvex)**.** *Under Assumptions A1 to A4 and $0 < \alpha \le 1/L$, for every $K \ge 1$,*

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\|\nabla f(x_k)\|^2 \le \frac{2\Delta}{\alpha K} + L\alpha\sigma^2, \qquad \Delta = f(x_0) - f^\star. \tag{3.2}$$

Proof. Taking the full expectation of Lemma 3.1 (tower property (R3)) gives $\mathbb{E}f(x_{k+1}) \le \mathbb{E}f(x_k) - \frac{\alpha}{2}\mathbb{E}\|\nabla f(x_k)\|^2 + \frac{L\alpha^2}{2}\sigma^2$. Summing over $k = 0, \dots, K-1$, the function values telescope:

$$\mathbb{E}f(x_K) \le f(x_0) - \frac{\alpha}{2}\sum_{k=0}^{K-1}\mathbb{E}\|\nabla f(x_k)\|^2 + K\frac{L\alpha^2}{2}\sigma^2.$$

Since $f(x_K) \ge f^\star$, rearranging gives $\frac{\alpha}{2}\sum_{k<K}\mathbb{E}\|\nabla f(x_k)\|^2 \le \Delta + K\frac{L\alpha^2}{2}\sigma^2$; divide by $\alpha K/2$. □

The two terms in (3.2) pull in opposite directions: a large α makes the first term small and the second large. Choosing $\alpha = \min\{1/L,\ \sigma^{-1}\sqrt{2\Delta/(LK)}\}$ balances them and gives

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\|\nabla f(x_k)\|^2 \le \frac{2L\Delta}{K} + 2\sigma\sqrt{\frac{2L\Delta}{K}}, \tag{3.3}$$

the well-known $O(1/\sqrt{K})$ rate of SGD [5]. (This is the same computation as in Corollary 6.2 below.)

3.3 Functions satisfying the PL inequality

Theorem 3.3 (SGD, PL)**.** *Under Assumptions A1 to A4, the PL inequality (2.2) with $\mu \le L$, and $0 < \alpha \le 1/L$, for every $k \ge 0$,*

$$\mathbb{E}f(x_k) - f^\star \le (1-\alpha\mu)^k\Delta + \frac{L\alpha\sigma^2}{2\mu}.$$

Proof. Subtract $f^\star$ from Lemma 3.1 and use (2.2): $\mathbb{E}_k[f(x_{k+1})] - f^\star \le (1-\alpha\mu)(f(x_k) - f^\star) + \frac{L\alpha^2}{2}\sigma^2$. With $e_k := \mathbb{E}f(x_k) - f^\star$ and the tower property, $e_{k+1} \le (1-\alpha\mu)e_k + \frac{L\alpha^2}{2}\sigma^2$. Note $0 \le 1-\alpha\mu < 1$, because $\alpha\mu \le \mu/L \le 1$. Unrolling the recursion and using the geometric series (Lemma 2.1(f)),

$$e_k \le (1-\alpha\mu)^k e_0 + \frac{L\alpha^2\sigma^2}{2}\sum_{j=0}^{k-1}(1-\alpha\mu)^j \le (1-\alpha\mu)^k\Delta + \frac{L\alpha^2\sigma^2}{2}\cdot\frac{1}{\alpha\mu}.$$

□

3.4 The template

Both proofs have the same three ingredients:

(T1) a *one-step expected decrease* of a nonnegative quantity (here $f(x_k) - f^\star$), with progress $\propto \alpha\|\nabla f(x_k)\|^2$ and noise cost $\propto \alpha^2\sigma^2$;

(T2) a *lower bound* on that quantity (here $f(x_K) - f^\star \geq 0$);

(T3) *telescoping* (nonconvex case) or a *contraction* obtained from the PL inequality (PL case).

For SGDM, **(T1)** is the only step that requires a new idea. Once we have it (Theorem 5.3), **(T2)** and **(T3)** go through as above.

4 SGD with momentum

4.1 The algorithm and its equivalent forms

Algorithm 1: SGD with momentum (heavy-ball form)

Input: starting point $x_0 \in \mathbb{R}^d$, step size $\alpha > 0$, momentum $\beta \in [0, 1)$. Set $x_{-1} := x_0$.
For $k = 0, 1, 2, \ldots$:

1. draw a fresh sample ξ_k and compute $g_k = g(x_k; \xi_k)$;
2. update $x_{k+1} = x_k - \alpha\, g_k + \beta\,(x_k - x_{k-1})$.

Software libraries usually implement momentum through an auxiliary "buffer". The next proposition shows that the common variants are the same algorithm.

Proposition 4.1 (Three equivalent forms)**.** *Run the following three recursions with the same samples $\xi_0, \xi_1, \ldots$, the same x_0, and $\gamma := \alpha/(1-\beta)$:*

(i) heavy-ball form: $x_{k+1} = x_k - \alpha g_k + \beta(x_k - x_{k-1}), \quad x_{-1} = x_0;$

(ii) momentum-buffer form: $m_k = \beta m_{k-1} + g_k, \quad x_{k+1} = x_k - \alpha m_k, \quad m_{-1} = 0;$

(iii) averaging form: $\bar{m}_k = \beta \bar{m}_{k-1} + (1-\beta) g_k, \quad x_{k+1} = x_k - \gamma \bar{m}_k, \quad \bar{m}_{-1} = 0.$

They produce identical iterates $x_0, x_1, x_2, \ldots$ Moreover $m_k = \sum_{j=0}^{k} \beta^{k-j} g_j$ and $\bar{m}_k = (1-\beta) m_k$.

Proof. Unrolling (ii) by induction gives $m_k = g_k + \beta m_{k-1} = \sum_{j=0}^{k} \beta^{k-j} g_j$. Next, the iterates of (ii) satisfy the recursion (i): for $k \geq 1$,

$$x_{k+1} - x_k = -\alpha m_k = -\alpha(\beta m_{k-1} + g_k) = \beta(-\alpha m_{k-1}) - \alpha g_k = \beta(x_k - x_{k-1}) - \alpha g_k,$$

and for $k = 0$, $x_1 - x_0 = -\alpha m_0 = -\alpha g_0 = \beta(x_0 - x_{-1}) - \alpha g_0$ because $x_{-1} = x_0$. Both recursions start at x_0; if their iterates agree up to index k, then $g_k = g(x_k; \xi_k)$ agrees too, and so does x_{k+1}. By induction the iterates coincide. For (iii), $\bar{m}_k = (1-\beta) m_k$ holds for $k = -1$ and is preserved by the recursions (multiply the recursion for m_k by $1-\beta$); hence $\gamma \bar{m}_k = \gamma(1-\beta) m_k = \alpha m_k$. □

Form (ii) is, for instance, PyTorch's momentum SGD, with learning rate α. Form (iii) is the "exponential moving average" form (used, e.g., for the first moment in Adam); there the learning rate *is* the effective step size γ introduced next. When comparing papers or code, always check

which form is meant: the same symbol "learning rate" can mean α or $\gamma = \alpha/(1-\beta)$, a factor of 10 apart when $\beta = 0.9$.

4.2 The effective step size

By Proposition 4.1, the displacement at step k is $-\alpha m_k = -\alpha \sum_{j \le k} \beta^{k-j} g_j$, an exponentially weighted sum of all past stochastic gradients. If the gradients were all equal to some vector g, this would be

$$-\alpha \frac{1-\beta^{k+1}}{1-\beta}\, g \;\longrightarrow\; -\frac{\alpha}{1-\beta}\, g = -\gamma\, g.$$

So in a region where the gradient changes slowly, SGDM moves like SGD with step size

$$\gamma := \frac{\alpha}{1-\beta} \qquad \text{(effective step size)}. \tag{4.1}$$

With $\beta = 0.9$, $\gamma = 10\alpha$. The effective step size will appear in every result below, and it is the right quantity for comparing SGDM with SGD (Section 8).

4.3 Why the function value is the wrong yardstick

Let us try to copy the proof of Lemma 3.1. Define the *velocity*

$$v_k := x_k - x_{k-1} \quad (k \ge 0), \qquad v_0 = 0, \qquad \text{so that} \qquad v_{k+1} = \beta v_k - \alpha g_k. \tag{4.2}$$

The descent lemma with $x = x_k$, $y = x_{k+1} = x_k + v_{k+1}$ gives $f(x_{k+1}) \le f(x_k) + \langle \nabla f(x_k),\, \beta v_k - \alpha g_k \rangle + \frac{L}{2}\|\beta v_k - \alpha g_k\|^2$. Since v_k is known at time k, taking $\mathbb{E}_k$ yields

$$\mathbb{E}_k[f(x_{k+1})] \le f(x_k) - \alpha \|\nabla f(x_k)\|^2 + \underbrace{\beta \langle \nabla f(x_k),\, v_k \rangle}_{\text{inertia term}} + \frac{L}{2}\, \mathbb{E}_k \|\beta v_k - \alpha g_k\|^2. \tag{4.3}$$

Compared with Lemma 3.1 there is a new *inertia term*. It is a problem for two reasons. First, it is linear in the velocity, and the velocity is itself of the order of the step size ($v_k = -\alpha m_{k-1}$), so the inertia term is of the *same order* as the progress term $-\alpha\|\nabla f(x_k)\|^2$; making α small does not make it negligible. Second, its sign is arbitrary: after the ball overshoots the bottom of a valley, the last step v_k points uphill at x_k and the term is positive. The following example shows that this really happens.

Example 4.2 (Momentum pushes the iterate away from the minimizer)**.** Let $f(x) = \frac{1}{2}x^2$ on $\mathbb{R}$ (so $L = 1$, $x^\star = 0$, $f^\star = 0$), use exact gradients ($g_k = x_k$), $\alpha = 1$, $\beta = \frac{1}{2}$, and $x_{-1} = x_0 = 1$. The update becomes $x_{k+1} = x_k - x_k + \frac{1}{2}(x_k - x_{k-1}) = \frac{1}{2}(x_k - x_{k-1})$, and the iterates are

k	0	1	2	3	4	5	6	7
x_k	1	0	$-\frac{1}{2}$	$-\frac{1}{4}$	$\frac{1}{8}$	$\frac{3}{16}$	$\frac{1}{32}$	$-\frac{5}{64}$
$f(x_k)$	$\frac{1}{2}$	0	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{128}$	$\frac{9}{512}$	$\frac{1}{2048}$	$\frac{25}{8192}$

At $k = 1$ the iterate sits *exactly at the minimizer*, with zero gradient. Nevertheless its velocity $v_1 = -1$ carries it away: $f(x_2) = \frac{1}{8} > 0 = f(x_1)$. The function value goes up again from $k = 4$ to $k = 5$ and from $k = 6$ to $k = 7$. (The method still converges here, see Proposition 7.2.) Figure 1(b) shows the same non-monotone behaviour for $\beta = 0.9$ and a step size small enough for all our theorems to apply.

So no inequality of the form (1.4) can hold for SGDM in general. What *can* be said about one step, measured by f itself, is the following.

Proposition 4.3 (One SGDM step measured by f)**.** *Under Assumptions A1, A3 and A4, for every $k \geq 0$,*

$$\mathbb{E}_k[f(x_{k+1})] \leq f(x_k) - \alpha(1 - L\alpha)\|\nabla f(x_k)\|^2 + \beta\langle \nabla f(x_k),\, v_k\rangle + L\beta^2\|v_k\|^2 + \frac{L\alpha^2}{2}\sigma^2. \tag{4.4}$$

Consequently, if $\alpha \leq \frac{1}{2L}$ and the last step points sufficiently downhill,

$$\langle \nabla f(x_k),\, v_k\rangle \leq -L\beta\|v_k\|^2, \tag{4.5}$$

then $\mathbb{E}_k[f(x_{k+1})] \leq f(x_k) - \frac{\alpha}{2}\|\nabla f(x_k)\|^2 + \frac{L\alpha^2}{2}\sigma^2$: exactly the SGD inequality of Lemma 3.1.

Proof. By the variance decomposition (Lemma 2.8; the conditional mean of $\beta v_k - \alpha g_k$ is $\beta v_k - \alpha\nabla f(x_k)$) and Lemma 2.1(e),

$$\mathbb{E}_k\|\beta v_k - \alpha g_k\|^2 = \|\beta v_k - \alpha\nabla f(x_k)\|^2 + \alpha^2\mathbb{E}_k\|\varepsilon_k\|^2 \leq 2\beta^2\|v_k\|^2 + 2\alpha^2\|\nabla f(x_k)\|^2 + \alpha^2\sigma^2.$$

Multiply by $\frac{L}{2}$ and insert into (4.3); this gives (4.4). Under (4.5), $\beta\langle \nabla f(x_k),\, v_k\rangle + L\beta^2\|v_k\|^2 \leq 0$, and $1 - L\alpha \geq \frac{1}{2}$ when $\alpha \leq \frac{1}{2L}$. □

So the function value decreases in expectation just as for SGD as long as the momentum points sufficiently downhill in the sense of (4.5); after an overshoot, when $\langle \nabla f(x_k),\, v_k\rangle > 0$, no decrease can be guaranteed. The algorithm has no control over the sign of this inner product, so a guarantee that holds at *every* step needs a different quantity to measure progress.

4.4 The extrapolated point

Definition 4.4 (Extrapolated point)**.** For $k \geq 0$ let

$$z_k := x_k + \frac{\beta}{1-\beta}\, v_k = x_k + \frac{\beta}{1-\beta}\,(x_k - x_{k-1}). \tag{4.6}$$

Lemma 4.5 (The extrapolated points perform SGD)**.** *For SGDM (Algorithm 1), $z_0 = x_0$ and, for every $k \geq 0$,*

$$z_{k+1} = z_k - \gamma\, g_k, \qquad \gamma = \frac{\alpha}{1-\beta}. \tag{4.7}$$

Proof. $z_0 = x_0$ because $v_0 = 0$. Using $x_{k+1} - x_k = v_{k+1}$ and the velocity recursion $v_{k+1} = \beta v_k - \alpha g_k$ from (4.2),

$$z_{k+1} - z_k = v_{k+1} + \frac{\beta}{1-\beta}(v_{k+1} - v_k) = \frac{1}{1-\beta}(v_{k+1} - \beta v_k) = \frac{-\alpha g_k}{1-\beta} = -\gamma g_k. \qquad \square$$

Physical meaning of z_k

Suppose that from time k on the force is switched off (all gradients are zero). Then the velocity

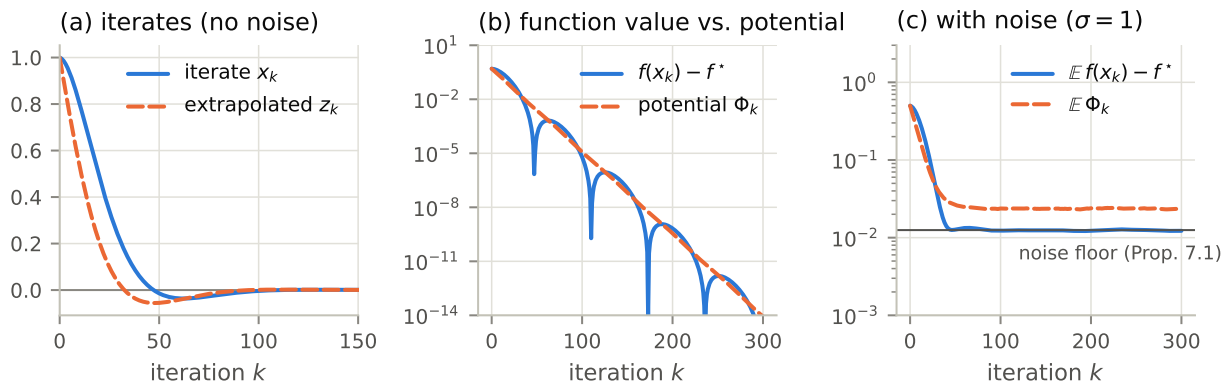


Figure 1: Heavy ball on $f(x) = \frac{1}{2}x^2$ ($L = 1$) with $\beta = 0.9$ and $\alpha = (1-\beta)^2/(2L) = 0.005$, the largest step allowed by Theorem 5.3(b). (a) Without noise: the iterate x_k and the extrapolated point z_k, which moves ahead of x_k. (b) Without noise: $f(x_k) - f^\star$ is not monotone (it rises after each overshoot), while the potential Φ_k of Theorem 5.3 decreases at every step. (c) With noise ($\sigma = 1$, average over 20 000 runs): both expectations settle at a noise floor; the floor of $\mathbb{E}f(x_k)$ agrees with the exact value of Proposition 7.1 (horizontal line).

decays geometrically, $v_{k+j} = \beta^j v_k$, and the ball comes to rest at

$$x_k + (\beta + \beta^2 + \beta^3 + \cdots)\, v_k = x_k + \frac{\beta}{1-\beta}\, v_k = z_k.$$

So z_k is the *coasting endpoint*: where the ball would stop under friction alone. A gradient kick $-\alpha g_k$ changes the velocity, and friction carries this change forward with total amplification $1 + \beta + \beta^2 + \cdots = \frac{1}{1-\beta}$. Hence the coasting endpoint moves by exactly $-\gamma g_k$: *the points z_k perform plain SGD with step size γ*—except that the gradient is evaluated at x_k rather than at z_k. The mismatch is controlled by the distance

$$\|z_k - x_k\| = \frac{\beta}{1-\beta}\, \|v_k\|, \tag{4.8}$$

which is proportional to the velocity. Figure 1(a) shows x_k and z_k for a one-dimensional quadratic: z_k "looks ahead" of x_k.

Remark 4.6 (Varying step sizes). If Algorithm 1 uses a step size α_k at iteration k (heavy-ball form, constant β), then $v_{k+1} = \beta v_k - \alpha_k g_k$, and the same computation gives $z_{k+1} = z_k - \gamma_k g_k$ with $\gamma_k = \alpha_k/(1-\beta)$. (In the momentum-buffer form a changing α_k rescales the whole buffer, the forms are no longer equivalent, and this identity fails; we only treat varying steps in heavy-ball form.)

5 The expected decrease inequality for SGDM

We now carry out step **(T1)** of the template for SGDM. The plan has three steps: (1) run the SGD argument on the extrapolated points z_k, which do perform SGD steps (Lemma 4.5), and pay for the fact that the gradient is evaluated at the "wrong" point x_k; (2) show that the velocity, which controls this payment, is damped by friction; (3) combine the two in a potential.

5.1 Step 1: descent at the extrapolated point

Lemma 5.1 (Descent at the extrapolated point)**.** *Under Assumptions A1, A3 and A4, for every* $k \geq 0$,

$$\mathbb{E}_k[f(z_{k+1})] \leq f(z_k) - \frac{\gamma}{2}(1 - L\gamma)\|\nabla f(x_k)\|^2 + \frac{L^2\gamma}{2}\|z_k - x_k\|^2 + \frac{L\gamma^2}{2}\sigma^2. \tag{5.1}$$

Proof. Descent lemma. By Lemma 4.5, $z_{k+1} = z_k - \gamma g_k$. Apply Lemma 2.3 with $x = z_k$ and $y = z_{k+1}$:

$$f(z_{k+1}) \leq f(z_k) - \gamma\langle \nabla f(z_k),\, g_k\rangle + \frac{L\gamma^2}{2}\|g_k\|^2.$$

Conditional expectation. The point z_k is known at time k (it is computed from x_k and x_{k-1}), hence so is $\nabla f(z_k)$. We take $\mathbb{E}_k$ of both sides (R4). By (R1), (R2) and Assumption A3, $\mathbb{E}_k\langle \nabla f(z_k),\, g_k\rangle = \langle \nabla f(z_k),\, \nabla f(x_k)\rangle$, and by Lemma 2.8, $\mathbb{E}_k\|g_k\|^2 \leq \|\nabla f(x_k)\|^2 + \sigma^2$. Therefore

$$\mathbb{E}_k[f(z_{k+1})] \leq f(z_k) - \gamma\langle \nabla f(z_k),\, \nabla f(x_k)\rangle + \frac{L\gamma^2}{2}(\|\nabla f(x_k)\|^2 + \sigma^2).$$

The gradient is taken at the wrong point. By polarization (Lemma 2.1(c)) with $a = \nabla f(z_k)$, $b = \nabla f(x_k)$, and then L-smoothness,

$$\begin{aligned} -\langle \nabla f(z_k),\, \nabla f(x_k)\rangle &= \tfrac{1}{2}\|\nabla f(z_k) - \nabla f(x_k)\|^2 - \tfrac{1}{2}\|\nabla f(z_k)\|^2 - \tfrac{1}{2}\|\nabla f(x_k)\|^2 \\ &\leq \tfrac{L^2}{2}\|z_k - x_k\|^2 - \tfrac{1}{2}\|\nabla f(x_k)\|^2, \end{aligned}$$

where we also dropped $-\frac{1}{2}\|\nabla f(z_k)\|^2 \leq 0$. Multiply by γ and insert into the previous display. □

Compared with SGD (Lemma 3.1), the essential new term is $\frac{L^2\gamma}{2}\|z_k - x_k\|^2$, the price of evaluating the gradient at x_k instead of z_k. (The progress coefficient is also somewhat smaller, $\frac{\gamma}{2}(1 - L\gamma)$ instead of $\gamma(1 - \frac{L\gamma}{2})$, because we discarded $-\frac{\gamma}{2}\|\nabla f(z_k)\|^2$.) By (4.8) it equals $\frac{L^2\gamma}{2}\big(\frac{\beta}{1-\beta}\big)^2\|v_k\|^2$: it is *quadratic* in the velocity, hence of higher order in the step size than the troublesome inertia term $\beta\langle \nabla f(x_k),\, v_k\rangle$ of (4.3), which was linear. This is what the change of variables $x_k \mapsto z_k$ buys.

5.2 Step 2: friction damps the velocity

Lemma 5.2 (Velocity recursion)**.** *Under Assumptions A3 and A4, for every* $k \geq 0$,

$$\mathbb{E}_k\|v_{k+1}\|^2 \leq \beta\|v_k\|^2 + (1 - \beta)\gamma^2\|\nabla f(x_k)\|^2 + (1 - \beta)^2\gamma^2\sigma^2. \tag{5.2}$$

Proof. By (4.2), $v_{k+1} = \beta v_k - \alpha g_k$. Since v_k is known at time k, its conditional mean is $\mathbb{E}_k[v_{k+1}] = \beta v_k - \alpha\nabla f(x_k)$ (rules (R1), (R2) and Assumption A3), and $v_{k+1} - \mathbb{E}_k[v_{k+1}] = -\alpha(g_k - \nabla f(x_k))$. The variance decomposition (Lemma 2.8) and Assumption A4 give

$$\mathbb{E}_k\|v_{k+1}\|^2 = \|\beta v_k - \alpha\nabla f(x_k)\|^2 + \alpha^2\mathbb{E}_k\|g_k - \nabla f(x_k)\|^2 \leq \|\beta v_k - \alpha\nabla f(x_k)\|^2 + \alpha^2\sigma^2.$$

Because $\alpha = (1 - \beta)\gamma$, we can write $\beta v_k - \alpha\nabla f(x_k) = \beta\, v_k + (1 - \beta)\,(-\gamma\nabla f(x_k))$, a convex combination. By Lemma 2.1(d) with $\theta = \beta$, $\|\beta v_k - \alpha\nabla f(x_k)\|^2 \leq \beta\|v_k\|^2 + (1 - \beta)\gamma^2\|\nabla f(x_k)\|^2$. Finally $\alpha^2 = (1 - \beta)^2\gamma^2$. □

In words: each step, friction keeps only a fraction β of the squared velocity, and the gradient and the noise inject new velocity.

5.3 Step 3: the potential and the main inequality

Theorem 5.3 (Expected decrease for SGDM)**.** *Let Assumptions A1 to A4 hold, $\beta \in [0,1)$, $\alpha > 0$, $\gamma = \alpha/(1-\beta)$, and define*

$$\Phi_k := f(z_k) - f^\star + c_\beta \|x_k - x_{k-1}\|^2 \ \geq 0, \qquad c_\beta := \frac{L\beta^2}{2(1-\beta)^2}. \tag{5.3}$$

(a) *For every $k \geq 0$ (no condition on α),*

$$\mathbb{E}_k[\Phi_{k+1}] \leq \Phi_k - \frac{\gamma}{2}\Big(1 - \frac{L\gamma\,(1-\beta+\beta^2)}{1-\beta}\Big)\|\nabla f(x_k)\|^2 - c_\beta\,(1-\beta-L\gamma)\|v_k\|^2 + \frac{1+\beta^2}{2}\,L\gamma^2\sigma^2. \tag{5.4}$$

(b) *If $\alpha \leq \frac{(1-\beta)^2}{2L}$, equivalently $L\gamma \leq \frac{1-\beta}{2}$, then for every $k \geq 0$,*

$$\mathbb{E}_k[\Phi_{k+1}] \leq \Phi_k - \frac{\gamma}{4}\|\nabla f(x_k)\|^2 - \frac{L\beta^2}{4(1-\beta)}\|x_k - x_{k-1}\|^2 + L\gamma^2\sigma^2. \tag{5.5}$$

Proof. By (4.8), $\frac{L^2\gamma}{2}\|z_k - x_k\|^2 = \frac{L^2\gamma}{2}\cdot\frac{\beta^2}{(1-\beta)^2}\|v_k\|^2 = L\gamma\,c_\beta\|v_k\|^2$. Subtracting $f^\star$ from Lemma 5.1 therefore gives

$$\mathbb{E}_k[f(z_{k+1})] - f^\star \leq f(z_k) - f^\star - \frac{\gamma}{2}(1-L\gamma)\|\nabla f(x_k)\|^2 + L\gamma\,c_\beta\|v_k\|^2 + \frac{L\gamma^2}{2}\sigma^2. \tag{5.6}$$

Multiplying Lemma 5.2 by $c_\beta \geq 0$ gives

$$c_\beta\,\mathbb{E}_k\|v_{k+1}\|^2 \leq c_\beta\beta\|v_k\|^2 + c_\beta(1-\beta)\gamma^2\|\nabla f(x_k)\|^2 + c_\beta(1-\beta)^2\gamma^2\sigma^2. \tag{5.7}$$

By linearity (R1), the sum of the left-hand sides is $\mathbb{E}_k[\Phi_{k+1}]$. Adding (5.6) and (5.7) and grouping terms,

$$\mathbb{E}_k[\Phi_{k+1}] \leq f(z_k) - f^\star + c_\beta(\beta+L\gamma)\|v_k\|^2 - \Big[\frac{\gamma}{2}(1-L\gamma) - c_\beta(1-\beta)\gamma^2\Big]\|\nabla f(x_k)\|^2 + \Big[\frac{L\gamma^2}{2} + c_\beta(1-\beta)^2\gamma^2\Big]\sigma^2.$$

We simplify the three coefficients using $c_\beta = \frac{L\beta^2}{2(1-\beta)^2}$:

- *velocity:* $c_\beta(\beta+L\gamma) = c_\beta - c_\beta(1-\beta-L\gamma)$, so $f(z_k) - f^\star + c_\beta(\beta+L\gamma)\|v_k\|^2 = \Phi_k - c_\beta(1-\beta-L\gamma)\|v_k\|^2$;
- *gradient:* $c_\beta(1-\beta)\gamma^2 = \frac{L\beta^2\gamma^2}{2(1-\beta)}$, so the bracket equals $\frac{\gamma}{2}(1 - L\gamma - \frac{L\gamma\beta^2}{1-\beta}) = \frac{\gamma}{2}(1 - \frac{L\gamma(1-\beta+\beta^2)}{1-\beta})$;
- *noise:* $c_\beta(1-\beta)^2\gamma^2 = \frac{L\beta^2\gamma^2}{2}$, so the bracket equals $\frac{1+\beta^2}{2}L\gamma^2$.

This proves (a). For (b), assume $L\gamma \leq \frac{1-\beta}{2}$. Then:

- $\frac{L\gamma(1-\beta+\beta^2)}{1-\beta} \leq \frac{1-\beta+\beta^2}{2} \leq \frac{1}{2}$ (because $\beta^2 \leq \beta$ for $\beta \in [0,1)$), so the gradient coefficient in (a) is at least $\frac{\gamma}{2}\cdot\frac{1}{2} = \frac{\gamma}{4}$;
- $1 - \beta - L\gamma \geq \frac{1-\beta}{2}$, so $c_\beta(1-\beta-L\gamma) \geq c_\beta\frac{1-\beta}{2} = \frac{L\beta^2}{4(1-\beta)}$;
- $\frac{1+\beta^2}{2} \leq 1$.

Inserting these three bounds into (5.4) gives (5.5). Finally, $\Phi_k \geq 0$ because $f(z_k) \geq f^\star$ (Assumption A2). □

5.4 Discussion

Side by side with SGD. Table 2 compares Theorem 5.3(b) with Lemma 3.1. The structure is identical; SGDM measures progress with Φ_k instead of $f(x_k) - f^\star$, and α is replaced by the effective step size γ. Both quantities start at the same value, $\Phi_0 = f(x_0) - f^\star = \Delta$, because $v_0 = 0$ and $z_0 = x_0$.

	SGD (Lemma 3.1)	SGDM (Theorem 5.3(b))
measure of progress	$f(x_k) - f^\star$	$\Phi_k = f(z_k) - f^\star + c_\beta \|x_k - x_{k-1}\|^2$
step-size condition	$\alpha \le 1/L$	$\alpha \le (1-\beta)^2/(2L)$, i.e. $\gamma \le (1-\beta)/(2L)$
guaranteed progress	$\frac{\alpha}{2}\|\nabla f(x_k)\|^2$	$\frac{\gamma}{4}\|\nabla f(x_k)\|^2$ (+ a velocity term)
price of noise	$\frac{L}{2}\alpha^2\sigma^2$	$L\gamma^2\sigma^2$
value at $k = 0$	Δ	Δ

Table 2: The expected decrease inequalities for SGD and SGDM.

Where the step-size condition comes from: an energy budget

The potential contains a kinetic energy $c_\beta\|v_k\|^2$. In each step friction removes the fraction $1-\beta$ of it (Lemma 5.2 keeps only $\beta\|v_k\|^2$), which releases $c_\beta(1-\beta)\|v_k\|^2$. Evaluating the gradient at the wrong point costs $L\gamma\, c_\beta\|v_k\|^2$ (Lemma 5.1). The budget is in surplus exactly when

$$L\gamma < 1-\beta \qquad \Longleftrightarrow \qquad \alpha < \frac{(1-\beta)^2}{L}\,:$$

friction must dissipate kinetic energy faster than the gradient mismatch creates error. This is visible in (5.4): if $L\gamma < 1-\beta$, both the velocity coefficient and the gradient coefficient are positive (for the latter note that $\frac{L\gamma(1-\beta+\beta^2)}{1-\beta} < 1-\beta+\beta^2 \le 1$), so without noise (for $\beta > 0$) Φ_k strictly decreases at every step unless $\nabla f(x_k) = 0$ and $v_k = 0$. Condition (b) keeps half of the friction as a safety margin, which produces the clean constants. As $\beta \to 1$ the friction vanishes, and so does the admissible step.

Remark 5.4 (The case $\beta = 0$). For $\beta = 0$ we have $c_\beta = 0$, $z_k = x_k$, $\Phi_k = f(x_k) - f^\star$, and (5.4) reads $\mathbb{E}_k[f(x_{k+1})] \le f(x_k) - \frac{\gamma}{2}(1-L\gamma)\|\nabla f(x_k)\|^2 + \frac{L\gamma^2}{2}\sigma^2$: this is Lemma 3.1 with a progress term smaller by about a factor 2, the price of discarding $-\frac{1}{2}\|\nabla f(z_k)\|^2$ in the polarization step. Constants in this report are chosen for simplicity and are not optimized.

Remark 5.5 (Why both parts of the potential are needed). Without the kinetic term, (5.6) leaves a positive term $L\gamma c_\beta\|v_k\|^2$ that $f(z_k)$ cannot absorb; adding $c_\beta\|v_k\|^2$ and Lemma 5.2 turns it into a decrease. Conversely, the mechanical energy $V_k = f(x_k) + \frac{\beta}{2\alpha}\|x_k - x_{k-1}\|^2$ of Proposition 7.3, which is built on $f(x_k)$, absorbs the first-order inertia term of (4.3) without noise; with noise, however, it picks up a noise term of first order in α (Remark 7.4).

Corollary 5.6 (When does one step decrease the potential?)**.** *Under the assumptions of Theorem 5.3(b),* $\mathbb{E}_k[\Phi_{k+1}] < \Phi_k$ *whenever*

$$\frac{\gamma}{4}\|\nabla f(x_k)\|^2 + \frac{L\beta^2}{4(1-\beta)}\|x_k - x_{k-1}\|^2 > L\gamma^2\sigma^2, \qquad \textit{in particular whenever} \qquad \|\nabla f(x_k)\|^2 > 4L\gamma\sigma^2.$$

Without noise ($\sigma = 0$), Φ_k *is non-increasing along every trajectory. For comparison, SGD with* $\alpha \le 1/L$ *decreases* f *in expectation whenever* $\|\nabla f(x_k)\|^2 > L\alpha\sigma^2$.

Proof. Immediate from (5.5) (and from Lemma 3.1 for SGD). □

So the circumstances under which SGDM makes progress *in a single step* are the same as for SGD: it suffices that the signal $\|\nabla f(x_k)\|^2$ exceeds the noise level, here $4L\gamma\sigma^2$; outside this "noise region" the potential decreases in expectation, inside it the noise may push it up. Shrinking γ shrinks the noise region.

The potential also controls the function value at the actual iterate:

Lemma 5.7 (The potential controls $f(x_k)$)**.** *Under Assumptions A1 and A2, for all $x, z \in \mathbb{R}^d$,*

$$f(x) - f^\star \le 2(f(z) - f^\star) + L\|x - z\|^2. \tag{5.8}$$

Consequently the SGDM iterates satisfy $f(x_k) - f^\star \le 2\,\Phi_k$ for all $k \ge 0$.

Proof. By the descent lemma with base point z, and Young's inequality (Lemma 2.1(b) with $\rho = 1/L$),

$$f(x) \le f(z) + \langle \nabla f(z),\, x - z\rangle + \frac{L}{2}\|x - z\|^2 \le f(z) + \frac{1}{2L}\|\nabla f(z)\|^2 + L\|x - z\|^2.$$

By Lemma 2.4, $\frac{1}{2L}\|\nabla f(z)\|^2 \le f(z) - f^\star$; subtracting $f^\star$ gives (5.8). For the iterates take $x = x_k$, $z = z_k$: by (4.8), $L\|x_k - z_k\|^2 = \frac{L\beta^2}{(1-\beta)^2}\|v_k\|^2 = 2c_\beta\|v_k\|^2$, so $f(x_k) - f^\star \le 2(f(z_k) - f^\star) + 2c_\beta\|v_k\|^2 = 2\Phi_k$. □

Without noise and under Theorem 5.3(b), this gives $f(x_k) - f^\star \le 2\Phi_k \le 2\Phi_0 = 2\Delta$: the function value may go up (Figure 1(b)), but never above $f^\star + 2(f(x_0) - f^\star)$.

Remark 5.8 (Two extensions). (i) *Varying step sizes.* Suppose Algorithm 1 uses α_k at iteration k (heavy-ball form) and set $\gamma_k = \alpha_k/(1-\beta)$. Each ingredient of the proof of Theorem 5.3 is a statement about a single iteration—Lemma 4.5 holds with γ_k by Remark 4.6, and Lemmas 5.1 and 5.2 only involve step k—and the weight c_β does not depend on the step size. Hence (5.5) holds at iteration k with γ_k in place of γ whenever $L\gamma_k \le \frac{1-\beta}{2}$, with the *same* potential Φ_k.

(ii) *Noise that grows with the gradient.* If Assumption A4 is replaced by $\mathbb{E}_k\|g_k - \nabla f(x_k)\|^2 \le \sigma^2 + M\|\nabla f(x_k)\|^2$ for some $M \ge 0$, then (5.5) still holds provided $\alpha \le \frac{(1-\beta)^2}{2L(1+M)}$. Indeed, the only changes in the proof are $\mathbb{E}_k\|g_k\|^2 \le (1+M)\|\nabla f(x_k)\|^2 + \sigma^2$ in Lemma 5.1 and an extra $\alpha^2 M\|\nabla f(x_k)\|^2$ in Lemma 5.2; the gradient coefficient becomes $\frac{\gamma}{2}[1 - L\gamma(1+M) - \frac{L\gamma\beta^2}{1-\beta}(1 + (1-\beta)M)] \ge \frac{\gamma}{2}[1 - \frac{1-\beta}{2} - \frac{\beta^2}{2}] \ge \frac{\gamma}{4}$ when $L\gamma(1+M) \le \frac{1-\beta}{2}$, and the other coefficients are unchanged.

6 Convergence theorems for SGDM

With Theorem 5.3 in hand, steps **(T2)**–**(T3)** of the template (Section 3.4) go through essentially verbatim. Recall $\Delta = f(x_0) - f^\star$ and $\gamma = \alpha/(1-\beta)$.

6.1 Nonconvex functions, constant step size

Theorem 6.1 (SGDM, nonconvex)**.** *Let Assumptions A1 to A4 hold, $\beta \in [0,1)$ and $0 < \alpha \le \frac{(1-\beta)^2}{2L}$. Then for every $K \ge 1$,*

$$\frac{1}{K}\sum_{k=0}^{K-1} \mathbb{E}\|\nabla f(x_k)\|^2 \le \frac{4\Delta}{\gamma K} + 4L\gamma\sigma^2. \tag{6.1}$$

Proof. Drop the (non-positive) velocity term in (5.5) and take full expectations (tower property (R3)): $\mathbb{E}\Phi_{k+1} \le \mathbb{E}\Phi_k - \frac{\gamma}{4}\mathbb{E}\|\nabla f(x_k)\|^2 + L\gamma^2\sigma^2$. Summing over $k = 0, \dots, K-1$, the potentials telescope:

$$\mathbb{E}\Phi_K \le \Phi_0 - \frac{\gamma}{4}\sum_{k=0}^{K-1} \mathbb{E}\|\nabla f(x_k)\|^2 + KL\gamma^2\sigma^2.$$

Now $\Phi_0 = f(z_0) - f^\star + c_\beta\|v_0\|^2 = f(x_0) - f^\star = \Delta$ and $\mathbb{E}\Phi_K \ge 0$. Rearranging, $\frac{\gamma}{4}\sum_{k<K}\mathbb{E}\|\nabla f(x_k)\|^2 \le \Delta + KL\gamma^2\sigma^2$; divide by $\gamma K/4$. □

Since a minimum never exceeds an average, (6.1) also bounds $\min_{k<K}\mathbb{E}\|\nabla f(x_k)\|^2$. In practice one can output x_R with R drawn uniformly from $\{0, \dots, K-1\}$, independently of everything else; then $\mathbb{E}\|\nabla f(x_R)\|^2$ equals the left-hand side of (6.1) [5].

Corollary 6.2 (Step size and iteration complexity)**.** *Under the assumptions of Theorem 6.1, assume $\Delta > 0$, fix $K \ge 1$ and choose*

$$\gamma = \min\Big\{\frac{1-\beta}{2L},\ \frac{1}{\sigma}\sqrt{\frac{\Delta}{LK}}\Big\} \quad \Big(\gamma = \tfrac{1-\beta}{2L} \text{ if } \sigma = 0\Big), \qquad \alpha = (1-\beta)\gamma.$$

Then

$$\frac{1}{K}\sum_{k=0}^{K-1} \mathbb{E}\|\nabla f(x_k)\|^2 \le \frac{8L\Delta}{(1-\beta)K} + 8\sigma\sqrt{\frac{L\Delta}{K}}. \tag{6.2}$$

Consequently $\min_{k<K}\mathbb{E}\|\nabla f(x_k)\|^2 \le \varepsilon^2$ *as soon as* $K \ge \max\{\frac{16L\Delta}{(1-\beta)\varepsilon^2},\ \frac{256L\Delta\sigma^2}{\varepsilon^4}\}$.

Proof. The chosen γ satisfies $L\gamma \le \frac{1-\beta}{2}$, so Theorem 6.1 applies. Using $\frac{1}{\min\{a,b\}} \le \frac{1}{a} + \frac{1}{b}$ for $a, b > 0$,

$$\frac{4\Delta}{\gamma K} \le \frac{4\Delta}{K}\Big(\frac{2L}{1-\beta} + \sigma\sqrt{\frac{LK}{\Delta}}\Big) = \frac{8L\Delta}{(1-\beta)K} + 4\sigma\sqrt{\frac{L\Delta}{K}}, \qquad 4L\gamma\sigma^2 \le 4L\sigma^2 \cdot \frac{1}{\sigma}\sqrt{\frac{\Delta}{LK}} = 4\sigma\sqrt{\frac{L\Delta}{K}}.$$

Adding gives (6.2). For the complexity statement, each of the two terms in (6.2) is at most $\varepsilon^2/2$ under the stated condition on K. □

Comparison with SGD. For SGD the analogous choice gives $\frac{2L\Delta}{K} + 2\sigma\sqrt{\frac{2L\Delta}{K}}$ (Section 3). The dominant term for large K (or small ε), of order $\sigma\sqrt{L\Delta/K}$, is the same for SGD and SGDM up to a constant: *our upper bound for SGDM matches the one for SGD in the dominant term, and it shows no acceleration.* Momentum costs a factor $1/(1-\beta)$ in the noise-free term, which reflects the more conservative step-size condition.

Corollary 6.3 (Exact gradients)**.** *Let Assumptions A1 and A2 hold, $\sigma = 0$ (that is, $g_k = \nabla f(x_k)$), $\beta \in [0,1)$ and $\alpha = \frac{(1-\beta)^2}{2L}$. Then*

(i) *$\Phi_{k+1} \le \Phi_k$ for all k, and $f(x_k) \le f^\star + 2\Delta$ for all k;*

(ii) *$\sum_{k=0}^{\infty} \|\nabla f(x_k)\|^2 \le \frac{8L\Delta}{1-\beta}$; in particular $\nabla f(x_k) \to 0$;*

(iii) *$\min_{0\le k<K} \|\nabla f(x_k)\|^2 \le \frac{8L\Delta}{(1-\beta)K}$.*

Proof. Without noise everything is deterministic, and (5.5) with $\gamma = \frac{1-\beta}{2L}$ reads $\Phi_{k+1} \le \Phi_k - \frac{\gamma}{4}\|\nabla f(x_k)\|^2$. (i) follows, together with $f(x_k) - f^\star \le 2\Phi_k \le 2\Phi_0 = 2\Delta$ (Lemma 5.7). (ii) Summing, $\frac{\gamma}{4}\sum_{k<K}\|\nabla f(x_k)\|^2 \le \Phi_0 - \Phi_K \le \Delta$ for every K, so the series converges with sum at most $4\Delta/\gamma = 8L\Delta/(1-\beta)$, and its terms tend to zero. (iii) The minimum is at most the average, $\frac{1}{K}\cdot\frac{8L\Delta}{1-\beta}$. □

6.2 Nonconvex functions, diminishing step sizes

Theorem 6.4 (SGDM with diminishing steps)**.** *Let Assumptions A1 to A4 hold and $\beta \in [0,1)$. Run Algorithm 1 in heavy-ball form with step size α_k at iteration k, $x_{k+1} = x_k - \alpha_k g_k + \beta(x_k - x_{k-1})$, where $\gamma_k := \alpha_k/(1-\beta)$ satisfies $0 < \gamma_k \le \frac{1-\beta}{2L}$ for all k. Then for every $K \ge 1$,*

$$\sum_{k=0}^{K-1} \gamma_k \,\mathbb{E}\|\nabla f(x_k)\|^2 \le 4\Delta + 4L\sigma^2 \sum_{k=0}^{K-1} \gamma_k^2. \tag{6.3}$$

Consequently:

(a) $\min\limits_{0\le k<K} \mathbb{E}\|\nabla f(x_k)\|^2 \le \dfrac{4\Delta + 4L\sigma^2\sum_{k<K}\gamma_k^2}{\sum_{k<K}\gamma_k}$.

(b) (Robbins–Monro steps.) *If $\sum_k \gamma_k = \infty$ and $\sum_k \gamma_k^2 < \infty$, then $\min_{k<K}\mathbb{E}\|\nabla f(x_k)\|^2 \to 0$ as $K \to \infty$ and $\liminf_{k\to\infty}\mathbb{E}\|\nabla f(x_k)\|^2 = 0$. Moreover, with probability one, $\sum_k \gamma_k\|\nabla f(x_k)\|^2 < \infty$ and $\liminf_{k\to\infty}\|\nabla f(x_k)\| = 0$.*

(c) *If $\gamma_k = \gamma_0/\sqrt{k+1}$ with $\gamma_0 \le \frac{1-\beta}{2L}$, then $\min\limits_{0\le k<K}\mathbb{E}\|\nabla f(x_k)\|^2 \le \dfrac{4\Delta + 4L\sigma^2\gamma_0^2(1+\ln K)}{\gamma_0\sqrt{K}} = O\Big(\dfrac{\log K}{\sqrt{K}}\Big)$.*

Proof. By Remark 5.8(i), inequality (5.5) holds at every iteration with γ_k: $\mathbb{E}_k\Phi_{k+1} \le \Phi_k - \frac{\gamma_k}{4}\|\nabla f(x_k)\|^2 + L\gamma_k^2\sigma^2$. Taking expectations, summing over $k < K$, and using $\Phi_0 = \Delta$ and $\mathbb{E}\Phi_K \ge 0$ gives $\frac{1}{4}\sum_{k<K}\gamma_k\mathbb{E}\|\nabla f(x_k)\|^2 \le \Delta + L\sigma^2\sum_{k<K}\gamma_k^2$, which is (6.3).

(a) The left side of (6.3) is at least $\big(\min_{k<K}\mathbb{E}\|\nabla f(x_k)\|^2\big)\sum_{k<K}\gamma_k$.

(b) Let $C := 4\Delta + 4L\sigma^2\sum_{k=0}^{\infty}\gamma_k^2 < \infty$. By (a), $\min_{k<K}\mathbb{E}\|\nabla f(x_k)\|^2 \le C/\sum_{k<K}\gamma_k \to 0$. If $\liminf_k \mathbb{E}\|\nabla f(x_k)\|^2$ were positive, there would be $\varepsilon > 0$ and k_0 with $\mathbb{E}\|\nabla f(x_k)\|^2 \ge \varepsilon$ for all $k \ge k_0$, and then the left side of (6.3) would be at least $\varepsilon\sum_{k_0\le k<K}\gamma_k \to \infty$, contradicting the bound C. For the almost-sure statement, the random variables $S_K := \sum_{k<K}\gamma_k\|\nabla f(x_k)\|^2$ are non-negative and non-decreasing in K, so $S_\infty := \lim_K S_K \in [0,\infty]$ exists, and by the monotone convergence theorem $\mathbb{E}S_\infty = \lim_K \mathbb{E}S_K \le C < \infty$. A non-negative random variable with finite expectation is finite with probability one, so $S_\infty < \infty$ almost surely. On this event $\liminf_k\|\nabla f(x_k)\| = 0$: otherwise $\|\nabla f(x_k)\|^2 \ge \varepsilon$ for some $\varepsilon > 0$ and all large k, which would force $S_\infty = \infty$ because $\sum_k \gamma_k = \infty$.

(c) $\sum_{k<K}\gamma_k = \gamma_0\sum_{j=1}^{K} j^{-1/2} \ge \gamma_0\sqrt{K}$, since each of the K terms is at least $K^{-1/2}$; and $\sum_{k<K}\gamma_k^2 = \gamma_0^2\sum_{j=1}^{K}\frac{1}{j} \le \gamma_0^2(1+\int_1^K \frac{dt}{t}) = \gamma_0^2(1+\ln K)$. Insert both into (a). □

Step sizes such as $\gamma_k = \gamma_0/(k+1)^{0.6}$ satisfy the Robbins–Monro conditions [12]. Stronger almost-sure statements (a limit rather than a lim inf) are known for SGD [1], and for the stochastic heavy-ball method in the convex case [14] or under additional assumptions and a different parametrization of the momentum [3]; they require more advanced martingale tools such as the Robbins–Siegmund theorem [13].

6.3 Functions satisfying the PL inequality

Lemma 6.5. *Under Assumptions A1 and A2 and the PL inequality (2.2) with $\mu \le L$, the SGDM iterates satisfy*

$$\Phi_k \le \frac{1}{\mu}\|\nabla f(x_k)\|^2 + 3c_\beta\|x_k - x_{k-1}\|^2 \qquad \text{for all } k \ge 0.$$

Proof. Write $f(z_k) - f^\star = (f(z_k) - f(x_k)) + (f(x_k) - f^\star)$. By the descent lemma and Young's inequality with $\rho = 1/L$, $f(z_k) - f(x_k) \le \langle \nabla f(x_k),\, z_k - x_k\rangle + \frac{L}{2}\|z_k - x_k\|^2 \le \frac{1}{2L}\|\nabla f(x_k)\|^2 + L\|z_k - x_k\|^2$, and by PL, $f(x_k) - f^\star \le \frac{1}{2\mu}\|\nabla f(x_k)\|^2$. Since $\frac{1}{2L} + \frac{1}{2\mu} \le \frac{1}{\mu}$ and $L\|z_k - x_k\|^2 = 2c_\beta\|v_k\|^2$ (see the proof of Lemma 5.7), $f(z_k) - f^\star \le \frac{1}{\mu}\|\nabla f(x_k)\|^2 + 2c_\beta\|v_k\|^2$. Add $c_\beta\|v_k\|^2$. □

Theorem 6.6 (SGDM under the PL inequality)**.** *Let Assumptions A1 to A4 and the PL inequality (2.2) with $\mu \le L$ hold, $\beta \in [0,1)$ and $0 < \alpha \le \frac{(1-\beta)^2}{2L}$. Then for all $k \ge 0$,*

$$\mathbb{E}\Phi_k \le \Big(1 - \frac{\gamma\mu}{4}\Big)^k\Delta + \frac{4L\gamma\sigma^2}{\mu}, \qquad \mathbb{E}f(x_k) - f^\star \le 2\Big(1 - \frac{\gamma\mu}{4}\Big)^k\Delta + \frac{8L\gamma\sigma^2}{\mu}.$$

Proof. Since $\frac{L\beta^2}{4(1-\beta)} = \frac{c_\beta(1-\beta)}{2}$, (5.5) reads

$$\mathbb{E}_k\Phi_{k+1} \le \Phi_k - \Big[\frac{\gamma}{4}\|\nabla f(x_k)\|^2 + \frac{c_\beta(1-\beta)}{2}\|v_k\|^2\Big] + L\gamma^2\sigma^2.$$

By Lemma 6.5, $\frac{\gamma\mu}{4}\Phi_k \le \frac{\gamma}{4}\|\nabla f(x_k)\|^2 + \frac{3\gamma\mu}{4}c_\beta\|v_k\|^2$, and $\frac{3\gamma\mu}{4}c_\beta \le \frac{c_\beta(1-\beta)}{2}$ because $\gamma\mu \le \gamma L \le \frac{1-\beta}{2} \le \frac{2(1-\beta)}{3}$. Hence the bracket is at least $\frac{\gamma\mu}{4}\Phi_k$, and

$$\mathbb{E}_k\Phi_{k+1} \le \Big(1 - \frac{\gamma\mu}{4}\Big)\Phi_k + L\gamma^2\sigma^2. \tag{6.4}$$

Note $0 < \frac{\gamma\mu}{4} \le \frac{1}{8}$. Taking expectations and unrolling as in the proof of Theorem 3.3, $\mathbb{E}\Phi_k \le (1-\frac{\gamma\mu}{4})^k\Phi_0 + L\gamma^2\sigma^2\sum_{j<k}(1-\frac{\gamma\mu}{4})^j \le (1-\frac{\gamma\mu}{4})^k\Delta + L\gamma^2\sigma^2\cdot\frac{4}{\gamma\mu}$. The bound on $f(x_k)$ follows from $f(x_k) - f^\star \le 2\Phi_k$ (Lemma 5.7). □

Compared with SGD (Theorem 3.3: contraction $1-\alpha\mu$, floor $\frac{L\alpha\sigma^2}{2\mu}$), the statement has the same form with α replaced by γ and worse constants. With the largest admissible step, $\gamma = \frac{1-\beta}{2L}$, the contraction factor is $1 - \frac{(1-\beta)\mu}{8L}$: *in this worst-case analysis momentum does not accelerate* (see Section 7.6).

Corollary 6.7 (PL inequality, decreasing steps)**.** *Under the assumptions of Theorem 6.6 (except for the step size), run Algorithm 1 in heavy-ball form with* $\gamma_k = \frac{8}{\mu(k+a)}$, $\alpha_k = (1-\beta)\gamma_k$, *where* $a \geq \frac{16L}{\mu(1-\beta)}$. *Then for all* $k \geq 0$,

$$\mathbb{E}\Phi_k \leq \frac{\nu}{k+a}, \qquad \mathbb{E}f(x_k) - f^\star \leq \frac{2\nu}{k+a}, \qquad \nu := \max\Big\{a\Delta,\ \frac{64L\sigma^2}{\mu^2}\Big\}.$$

Proof. Since $\gamma_k \leq \gamma_0 = \frac{8}{\mu a} \leq \frac{1-\beta}{2L}$, the one-step contraction (6.4) holds at every iteration with γ_k (its proof uses only (5.5) at iteration k, valid by Remark 5.8(i), and Lemma 6.5). With $e_k := \mathbb{E}\Phi_k$ and $\frac{\gamma_k \mu}{4} = \frac{2}{k+a}$,

$$e_{k+1} \leq \Big(1 - \frac{2}{k+a}\Big)e_k + \frac{64L\sigma^2}{\mu^2(k+a)^2}.$$

We show $e_k \leq \nu/(k+a)$ by induction. For $k=0$, $e_0 = \Delta \leq \nu/a$. If $e_k \leq \nu/(k+a)$, then, because $1 - \frac{2}{k+a} \geq 0$ (note $a \geq 16$ because $\mu \leq L$) and $\frac{64L\sigma^2}{\mu^2} \leq \nu$,

$$e_{k+1} \leq \frac{(k+a-2)\,\nu}{(k+a)^2} + \frac{\nu}{(k+a)^2} = \frac{(k+a-1)\,\nu}{(k+a)^2} \leq \frac{\nu}{k+a+1},$$

where the last step uses $(k+a-1)(k+a+1) = (k+a)^2 - 1 \leq (k+a)^2$. The bound on $f(x_k)$ follows from Lemma 5.7. □

6.4 Convex functions

For convex functions a different potential—a distance to the minimizer, as in the classical convex analysis of SGD—gives a guarantee under a step-size condition that does not degrade as $\beta \to 1$.

Theorem 6.8 (SGDM, convex)**.** *Let Assumptions A1, A3 and A4 hold, let* f *be convex with a minimizer* $x^\star$ *(so* $f^\star = f(x^\star)$*),* $\beta \in [0,1)$, *and* $0 < \alpha \leq \frac{1-\beta}{2L}$, *i.e.* $L\gamma \leq \frac{1}{2}$. *Define*

$$\Psi_k := \|z_k - x^\star\|^2 + \frac{2\gamma\beta}{1-\beta}(f(x_{k-1}) - f^\star) \ \geq 0.$$

(a) *For every* $k \geq 0$: $\mathbb{E}_k\Psi_{k+1} \leq \Psi_k - 2\gamma(1-L\gamma)(f(x_k)-f^\star) + \gamma^2\sigma^2 \leq \Psi_k - \gamma(f(x_k)-f^\star) + \gamma^2\sigma^2$.

(b) *For every* $K \geq 1$, *with* $\bar{x}_K := \frac{1}{K}\sum_{k<K} x_k$,

$$\mathbb{E}f(\bar{x}_K) - f^\star \leq \frac{1}{K}\sum_{k=0}^{K-1}(\mathbb{E}f(x_k) - f^\star) \leq \frac{\|x_0 - x^\star\|^2}{\gamma K} + \frac{2\beta\Delta}{(1-\beta)K} + \gamma\sigma^2.$$

(c) *With* $D := \|x_0 - x^\star\| > 0$ *and* $\gamma = \min\{\frac{1}{2L}, \frac{D}{\sigma\sqrt{K}}\}$ *(*$\gamma = \frac{1}{2L}$ *if* $\sigma = 0$*):* $\mathbb{E}f(\bar{x}_K) - f^\star \leq \frac{2LD^2}{K} + \frac{2\beta\Delta}{(1-\beta)K} + \frac{2\sigma D}{\sqrt{K}}$.

Proof. (a) By Lemma 4.5, $z_{k+1} - x^\star = (z_k - x^\star) - \gamma g_k$, hence $\|z_{k+1} - x^\star\|^2 = \|z_k - x^\star\|^2 - 2\gamma\langle g_k,\, z_k - x^\star\rangle + \gamma^2\|g_k\|^2$. Taking $\mathbb{E}_k$ (z_k is known at time k; Assumption A3; Lemma 2.8):

$$\mathbb{E}_k\|z_{k+1} - x^\star\|^2 \leq \|z_k - x^\star\|^2 - 2\gamma\langle\nabla f(x_k),\, z_k - x^\star\rangle + \gamma^2(\|\nabla f(x_k)\|^2 + \sigma^2).$$

Split $z_k - x^\star = (x_k - x^\star) + \frac{\beta}{1-\beta}(x_k - x_{k-1})$ and use convexity (Definition 2.5) twice: with $y = x^\star$, $\langle \nabla f(x_k), x_k - x^\star \rangle \geq f(x_k) - f^\star$; with $y = x_{k-1}$, $\langle \nabla f(x_k), x_k - x_{k-1} \rangle \geq f(x_k) - f(x_{k-1})$. Together with $\|\nabla f(x_k)\|^2 \leq 2L(f(x_k) - f^\star)$ (Lemma 2.4) this yields

$$\mathbb{E}_k \|z_{k+1} - x^\star\|^2 \leq \|z_k - x^\star\|^2 - 2\gamma(1 - L\gamma)(f(x_k) - f^\star) + \frac{2\gamma\beta}{1-\beta}(f(x_{k-1}) - f(x_k)) + \gamma^2\sigma^2.$$

Add $\frac{2\gamma\beta}{1-\beta}(f(x_k) - f^\star)$, which is known at time k, to both sides. The left side becomes $\mathbb{E}_k \Psi_{k+1}$ and the right side becomes $\Psi_k - 2\gamma(1 - L\gamma)(f(x_k) - f^\star) + \gamma^2\sigma^2$. Finally $L\gamma \leq \frac{1}{2}$ gives $2\gamma(1 - L\gamma) \geq \gamma$.

(b) Take expectations in (a) and sum over $k < K$: $\mathbb{E}\Psi_K \leq \Psi_0 - \gamma \sum_{k<K} (\mathbb{E} f(x_k) - f^\star) + K\gamma^2\sigma^2$, where $\Psi_0 = \|x_0 - x^\star\|^2 + \frac{2\gamma\beta}{1-\beta}\Delta$ (recall $z_0 = x_0 = x_{-1}$) and $\mathbb{E}\Psi_K \geq 0$. Rearrange and divide by γK. The first inequality is Jensen's inequality: by convexity, $f(x_k) \geq f(\bar{x}_K) + \langle \nabla f(\bar{x}_K), x_k - \bar{x}_K \rangle$ for each k, and averaging over k the inner products cancel.

(c) As in Corollary 6.2: $\frac{D^2}{\gamma K} \leq \frac{D^2}{K}(2L + \frac{\sigma\sqrt{K}}{D})$ and $\gamma\sigma^2 \leq \frac{\sigma D}{\sqrt{K}}$. □

Remark 6.9 (Why convexity relaxes the step-size condition). In the convex case the step-size restriction $L\gamma \leq \frac{1}{2}$ does not depend on β—it is the SGD condition with α replaced by γ—whereas the nonconvex analysis needs $L\gamma \leq \frac{1-\beta}{2}$. Convexity lets us bound the troublesome inner product $\langle \nabla f(x_k), x_k - x_{k-1} \rangle$ by a difference of function values, which telescopes, instead of paying for it with kinetic energy. The extra term $\frac{2\beta\Delta}{(1-\beta)K}$ is the price of momentum; since $\Delta \leq \frac{L}{2}D^2$ (descent lemma at $x^\star$, where $\nabla f(x^\star) = 0$), it is at most $\frac{\beta}{1-\beta} \cdot \frac{LD^2}{K}$.

7 Under what circumstances does SGDM converge?

The theorems of Section 6 hold under explicit *sufficient* conditions. This section asks which parts of these conditions can be removed, what goes wrong when they fail, and how they fit together. Quadratic functions, for which everything can be computed exactly, serve as test cases.

7.1 A constant step size leaves a noise floor

All constant-step results contain a term proportional to $\gamma\sigma^2$ that does not vanish as $K \to \infty$. The next proposition shows that this is not an artifact of the proofs: the floor is real, of exactly this order, and the same for SGD and SGDM at equal effective step size.

Proposition 7.1 (Exact noise floor on a quadratic). *Let $f(x) = \frac{h}{2}x^2$ on $\mathbb{R}$ with $h > 0$, and let $g_k = hx_k + \xi_k$, where $\xi_0, \xi_1, \ldots$ are independent with $\mathbb{E}\xi_k = 0$ and $\mathbb{E}\xi_k^2 = \sigma^2$ (so Assumptions A1 to A4 hold with $L = h$). Let $\beta \in [0, 1)$ and $0 < \alpha h < 2(1 + \beta)$. Then, for every starting point x_0,*

$$\lim_{k\to\infty} \mathbb{E} f(x_k) = \frac{\gamma\sigma^2}{2} \cdot \frac{1+\beta}{2(1+\beta) - \alpha h}, \qquad \lim_{k\to\infty} \mathbb{E}[f'(x_k)^2] = h\gamma\sigma^2 \cdot \frac{1+\beta}{2(1+\beta) - \alpha h}. \tag{7.1}$$

In particular $\lim_k \mathbb{E} f(x_k) \geq \frac{\gamma\sigma^2}{4}$, and $\lim_k \mathbb{E} f(x_k) = \frac{\gamma\sigma^2}{4} + \frac{\gamma\sigma^2}{4} \cdot \frac{\alpha h}{2(1+\beta)-\alpha h}$, so to first order the floor depends on (α, β) only through $\gamma = \alpha/(1 - \beta)$.

Proof. Let $a := 1 + \beta - \alpha h$; the update reads $x_{k+1} = ax_k - \beta x_{k-1} - \alpha\xi_k$. With $X_k := (x_k, x_{k-1})^\top$ (and $x_{-1} = x_0$),

$$X_{k+1} = MX_k + w_k, \qquad M := \begin{pmatrix} a & -\beta \\ 1 & 0 \end{pmatrix}, \qquad w_k := \begin{pmatrix} -\alpha\xi_k \\ 0 \end{pmatrix}.$$

Second moments. Let $C_k := \mathbb{E}[X_k X_k^\top]$, a symmetric 2×2 matrix whose $(1,1)$ entry is $\mathbb{E}x_k^2$. Since X_k is a function of $\xi_0,\dots,\xi_{k-1}$, it is independent of ξ_k, so $\mathbb{E}[X_k w_k^\top] = \mathbb{E}[X_k]\,\mathbb{E}[w_k]^\top = 0$. Expanding the product,

$$C_{k+1} = \mathbb{E}\big[(MX_k + w_k)(MX_k + w_k)^\top\big] = MC_kM^\top + Q, \qquad Q := \mathbb{E}[w_k w_k^\top] = \begin{pmatrix} \alpha^2\sigma^2 & 0 \\ 0 & 0 \end{pmatrix},$$

and by induction $C_k = M^k C_0 (M^\top)^k + \sum_{j=0}^{k-1} M^j Q (M^\top)^j$.

Convergence. The eigenvalues of M are the roots of $\det(\lambda I - M) = \lambda^2 - a\lambda + \beta$. By the proof of Proposition 7.2 below, the assumption $0 < \alpha h < 2(1+\beta)$ implies that both roots have modulus less than one. Then the entries of M^k tend to zero geometrically fast (Lemma A.2 in the appendix), so $M^k C_0 (M^\top)^k \to 0$ and the series converges: $C_k \to C_\infty := \sum_{j\ge 0} M^j Q (M^\top)^j$. Passing to the limit in the recursion, $C_\infty = MC_\infty M^\top + Q$.

Solving for the limit. Write $C_\infty = \begin{pmatrix} s & r \\ r & t \end{pmatrix}$. The second row of M is $(1,0)$ and the first is $(a,-\beta)$, so comparing entries of $C_\infty = MC_\infty M^\top + Q$ gives

$$(2,2):\ t = s; \qquad (1,2):\ r = as - \beta r; \qquad (1,1):\ s = a^2 s - 2a\beta r + \beta^2 t + \alpha^2\sigma^2.$$

Hence $r = \frac{as}{1+\beta}$, and substituting into the $(1,1)$ equation and multiplying by $1+\beta$: $s\big[(1+\beta)(1-\beta^2) - a^2(1+\beta) + 2a^2\beta\big] = \alpha^2\sigma^2(1+\beta)$. The bracket equals $(1+\beta)(1-\beta^2) - a^2(1-\beta) = (1-\beta)\big[(1+\beta)^2 - a^2\big]$, and $(1+\beta)^2 - a^2 = (1+\beta-a)(1+\beta+a) = \alpha h\,\big(2(1+\beta) - \alpha h\big)$. Therefore

$$s = \frac{\alpha^2\sigma^2(1+\beta)}{(1-\beta)\,\alpha h\,(2(1+\beta)-\alpha h)} = \frac{\gamma\sigma^2}{h}\cdot\frac{1+\beta}{2(1+\beta)-\alpha h}.$$

Since $\mathbb{E}f(x_k) = \frac{h}{2}\mathbb{E}x_k^2 \to \frac{h}{2}s$ and $\mathbb{E}f'(x_k)^2 = h^2\mathbb{E}x_k^2 \to h^2 s$, we obtain (7.1). Finally, $\frac{1+\beta}{2(1+\beta)-\alpha h} = \frac{1}{2} + \frac{\alpha h}{2(2(1+\beta)-\alpha h)} \ge \frac{1}{2}$. □

Consequences. (i) With a constant step and $\sigma > 0$, $\mathbb{E}f(x_k)$ does not converge to $f^\star$; the floor is of order $\gamma\sigma^2$, exactly as in Theorems 6.1 and 6.6. With $L = h$, (7.1) gives $\lim_k \mathbb{E}f'(x_k)^2 \approx \frac{L\gamma\sigma^2}{2}$, while Theorem 6.1 guarantees at most $4L\gamma\sigma^2$: the theorem is pessimistic only by a constant factor. (ii) At equal *effective* step size, SGD ($\beta = 0$) and SGDM have the same floor to first order: momentum does not average the noise away. At equal *raw* step α, momentum raises the floor by the factor $1/(1-\beta)$. (iii) To converge exactly, one must let the step size decrease (Theorem 6.4 and Corollary 6.7), reduce the variance (larger mini-batches, Remark 2.9), or have exact gradients (Corollary 6.3). Figure 5 confirms (7.1) numerically.

7.2 Too large a step size: the exact stability region on quadratics

Proposition 7.2 (Heavy ball on a quadratic)**.** *Let $f(x) = \frac{h}{2}x^2$ with $h > 0$, exact gradients, $\beta \in [0,1)$ and $\alpha > 0$. The heavy-ball iteration $x_{k+1} = (1+\beta-\alpha h)x_k - \beta x_{k-1}$ converges to 0 for every initial pair (x_{-1}, x_0) if and only if*

$$\alpha h < 2(1+\beta). \tag{7.2}$$

In that case the convergence is linear. More precisely, let $\lambda_{1,2}$ be the roots of $p(\lambda) = \lambda^2 - (1+\beta-\alpha h)\lambda + \beta$; then for every ρ with $\max\{|\lambda_1|,|\lambda_2|\} < \rho < 1$ there is a constant C (depending on ρ

and the initial values) such that $|x_k| \le C\rho^k$*, and if* $\lambda_1 \ne \lambda_2$ *one may take* $\rho = \max\{|\lambda_1|, |\lambda_2|\}$*. For* $f(x) = \frac{1}{2}x^\top Hx$ *with* H *symmetric positive definite with largest eigenvalue* L*, the iteration converges for all initial conditions if and only if* $\alpha L < 2(1+\beta)$*.*

Proof. Let $a = 1 + \beta - \alpha h$. By Lemma A.1 in the appendix, every solution of $x_{k+1} = ax_k - \beta x_{k-1}$ has the form $c_1\lambda_1^k + c_2\lambda_2^k$ (distinct roots) or $(c_1 + c_2 k)\lambda_1^k$ (double root), and every such sequence is a solution. Hence all solutions tend to 0 if both roots satisfy $|\lambda_i| < 1$; and if some root has $|\lambda_1| \ge 1$ and is real, the solution with $x_{-1} = 1$, $x_0 = \lambda_1$, namely $x_k = \lambda_1^{k+1}$, does not tend to 0. It remains to relate the roots to (7.2).

If (7.2) *holds, both roots have modulus* < 1*.* If the roots are complex ($a^2 < 4\beta$), they are conjugate, and $|\lambda_1|^2 = \lambda_1\lambda_2 = \beta < 1$. If they are real ($a^2 \ge 4\beta$), observe that $p(1) = 1 - a + \beta = \alpha h > 0$ and $p(-1) = 1 + a + \beta = 2(1+\beta) - \alpha h > 0$. Since p is an upward parabola, $p \le 0$ on the interval $[\lambda_1, \lambda_2]$ between its roots, so neither 1 nor -1 belongs to this interval; it therefore lies entirely in $(-\infty, -1)$, in $(-1, 1)$, or in $(1, \infty)$. In the first and third cases $|\lambda_1\lambda_2| > 1$, contradicting $\lambda_1\lambda_2 = \beta < 1$. Hence both roots lie in $(-1, 1)$.

If (7.2) *fails, some root is real with modulus* ≥ 1*.* Then $p(-1) = 2(1+\beta) - \alpha h \le 0$, and since $p(\lambda) \to +\infty$ as $\lambda \to -\infty$, the intermediate value theorem gives a real root $\lambda \le -1$.

For the matrix case, diagonalize $H = U\Lambda U^\top$ with U orthogonal. In the coordinates $y = U^\top x$ the iteration decouples into d scalar iterations with $h = \Lambda_{ii}$, and all of them converge for all initial conditions if and only if $\alpha\Lambda_{ii} < 2(1+\beta)$ for every i, i.e. $\alpha L < 2(1+\beta)$. □

Momentum on quadratics. For gradient descent ($\beta = 0$) the condition is the familiar $\alpha L < 2$. Momentum enlarges it to $\alpha L < 2(1+\beta)$, or, in terms of the effective step, $\gamma L < \frac{2(1+\beta)}{1-\beta}$, which equals 38 for $\beta = 0.9$. On quadratics momentum can also accelerate: if the eigenvalues of H lie in $[\mu, L]$ and $\kappa = L/\mu$, Polyak's choice [11] $\alpha = \frac{4}{(\sqrt{L}+\sqrt{\mu})^2}$, $\beta = \big(\frac{\sqrt{L}-\sqrt{\mu}}{\sqrt{L}+\sqrt{\mu}}\big)^2$ gives $(1-\sqrt{\beta})^2 = \alpha\mu$ and $(1+\sqrt{\beta})^2 = \alpha L$, so for every eigenvalue $h \in [\mu, L]$ we have $|1 + \beta - \alpha h| \le 2\sqrt{\beta}$; the roots are then complex (or double) with modulus $\sqrt{\beta} = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$, compared with the rate $\frac{\kappa-1}{\kappa+1}$ of optimally tuned gradient descent. Both facts are special to quadratics with exact gradients: Example 7.5 below shows that the same parameters can fail on a smooth strongly convex function, and Proposition 7.1 shows that with noise the floor is governed by γ.

7.3 Without noise, larger steps are safe for every smooth function

Proposition 7.3 (Mechanical energy decreases)**.** *Let Assumptions A1 and A2 hold, use exact gradients* ($g_k = \nabla f(x_k)$)*,* $\beta \in [0, 1)$ *and* $0 < \alpha < \frac{2(1-\beta)}{L}$*. Let* $V_k := f(x_k) + \frac{\beta}{2\alpha}\|x_k - x_{k-1}\|^2$*. Then for all* $k \ge 0$*,*

$$V_{k+1} \le V_k - \Big(\frac{1-\beta}{\alpha} - \frac{L}{2}\Big)\|x_{k+1} - x_k\|^2, \tag{7.3}$$

and $\min_{0\le k<K}\|\nabla f(x_k)\|^2 \le \frac{2(1+\beta^2)\Delta}{\alpha\,(1-\beta-\alpha L/2)\,K}$*. In particular* $\nabla f(x_k) \to 0$*.*

Proof. With $v_k = x_k - x_{k-1}$, the update gives $\alpha\nabla f(x_k) = \beta v_k - v_{k+1}$. The descent lemma with $x = x_k$, $y = x_{k+1}$ and Young's inequality ($\rho = 1$) give

$$f(x_{k+1}) \le f(x_k) + \frac{1}{\alpha}\big(\beta\langle v_k,\, v_{k+1}\rangle - \|v_{k+1}\|^2\big) + \frac{L}{2}\|v_{k+1}\|^2 \le f(x_k) + \frac{\beta}{2\alpha}\|v_k\|^2 + \Big(\frac{\beta}{2\alpha} - \frac{1}{\alpha} + \frac{L}{2}\Big)\|v_{k+1}\|^2.$$

Adding $\frac{\beta}{2\alpha}\|v_{k+1}\|^2$ to both sides gives (7.3). The coefficient $c := \frac{1-\beta}{\alpha} - \frac{L}{2}$ is positive by assumption. Summing (7.3) and using $V_0 = f(x_0)$ (as $v_0 = 0$) and $V_K \geq f^\star$, $\sum_{j=1}^K \|v_j\|^2 \leq \Delta/c$. Finally $\|\nabla f(x_k)\|^2 = \alpha^{-2}\|\beta v_k - v_{k+1}\|^2 \leq 2\alpha^{-2}(\beta^2\|v_k\|^2 + \|v_{k+1}\|^2)$ by Lemma 2.1(e); summing over $k < K$ and using $v_0 = 0$, $\sum_{k<K}\|\nabla f(x_k)\|^2 \leq \frac{2(1+\beta^2)}{\alpha^2}\sum_{j=1}^K \|v_j\|^2 \leq \frac{2(1+\beta^2)\Delta}{\alpha^2 c}$, and $\alpha^2 c = \alpha(1-\beta-\alpha L/2)$. The minimum is at most the average. □

This "kinetic plus potential energy" argument is classical [18]; see also [4] for the convex case. The condition $\alpha < \frac{2(1-\beta)}{L}$ reads $\gamma < \frac{2}{L}$: exactly the step-size range of gradient descent. So *without noise*, heavy ball is safe for every smooth function as long as its effective step is a safe gradient-descent step.

Remark 7.4 (Why the energy argument fails with noise). With $g_k = \nabla f(x_k) + \varepsilon_k$, the same computation gives $V_{k+1} \leq V_k - c\|v_{k+1}\|^2 - \langle \varepsilon_k, v_{k+1}\rangle$. Since $v_{k+1} = (\beta v_k - \alpha\nabla f(x_k)) - \alpha\varepsilon_k$, we get $\mathbb{E}_k\langle \varepsilon_k, v_{k+1}\rangle = -\alpha\mathbb{E}_k\|\varepsilon_k\|^2$ and $\mathbb{E}_k\|v_{k+1}\|^2 = \|\beta v_k - \alpha\nabla f(x_k)\|^2 + \alpha^2\mathbb{E}_k\|\varepsilon_k\|^2$, hence

$$\mathbb{E}_k V_{k+1} \leq V_k - c\,\|\beta v_k - \alpha\nabla f(x_k)\|^2 + \alpha\Big(\beta + \frac{\alpha L}{2}\Big)\mathbb{E}_k\|\varepsilon_k\|^2.$$

For $\beta = 0$ the noise term is the SGD term $\frac{L\alpha^2}{2}\sigma^2$, but for $\beta > 0$ it is of *first* order in α. Telescoped over K steps it contributes $K\alpha\beta\sigma^2$, which is of the same order as the total progress, and the resulting bound does not vanish as $\alpha \to 0$. The potential Φ_k of Theorem 5.3 pays only $L\gamma^2\sigma^2$ per step; this is why the stochastic analysis is built on the extrapolated point.

7.4 Quadratic intuition can mislead: a counterexample

Example 7.5 (Lessard–Recht–Packard [8])**.** Consider the piecewise quadratic function

$$f(x) = \begin{cases} \frac{25}{2}x^2, & x < 1, \\ \frac{1}{2}x^2 + 24x - 12, & 1 \leq x < 2, \\ \frac{25}{2}x^2 - 24x + 36, & x \geq 2, \end{cases} \qquad f'(x) = \begin{cases} 25x, & x < 1, \\ x + 24, & 1 \leq x < 2, \\ 25x - 24, & x \geq 2. \end{cases}$$

One checks that f and f' are continuous at $x = 1$ and $x = 2$, and that $f'' \in \{1, 25\}$ away from these points, so f is 1-strongly convex and 25-smooth with minimizer $x^\star = 0$. Take Polyak's quadratic-optimal parameters for $\mu = 1$, $L = 25$: $\alpha = \frac{4}{(5+1)^2} = \frac{1}{9}$ and $\beta = \big(\frac{5-1}{5+1}\big)^2 = \frac{4}{9}$. These satisfy (7.2) ($\alpha L = \frac{25}{9} < \frac{26}{9} = 2(1+\beta)$), so on *every* quadratic with curvature in $[1, 25]$ heavy ball converges, at rate $\frac{2}{3}$.

On f, however, it does not. With these parameters the update reads $x_{k+1} = \frac{13}{9}x_k - \frac{4}{9}x_{k-1} - \frac{1}{9}f'(x_k)$, that is,

$$x_{k+1} = -\tfrac{4}{3}x_k - \tfrac{4}{9}x_{k-1} \quad \text{if } x_k < 1, \qquad x_{k+1} = -\tfrac{4}{3}x_k - \tfrac{4}{9}x_{k-1} + \tfrac{8}{3} \quad \text{if } x_k \geq 2.$$

Let $p_1 = \frac{792}{1225} \approx 0.647$, $p_2 = -\frac{2208}{1225} \approx -1.802$ and $p_3 = \frac{2592}{1225} \approx 2.116$. Starting from $(x_{k-1}, x_k) = (p_3, p_1)$:

$$\begin{aligned} p_1 < 1: &\quad -\tfrac{4}{3}\cdot\tfrac{792}{1225} - \tfrac{4}{9}\cdot\tfrac{2592}{1225} = \tfrac{-1056-1152}{1225} = p_2, \\ p_2 < 1: &\quad -\tfrac{4}{3}\cdot\big(-\tfrac{2208}{1225}\big) - \tfrac{4}{9}\cdot\tfrac{792}{1225} = \tfrac{2944-352}{1225} = p_3, \\ p_3 \geq 2: &\quad -\tfrac{4}{3}\cdot\tfrac{2592}{1225} - \tfrac{4}{9}\cdot\big(-\tfrac{2208}{1225}\big) + \tfrac{8}{3} = \tfrac{-31104+8832+29400}{11025} = \tfrac{7128}{11025} = p_1. \end{aligned}$$

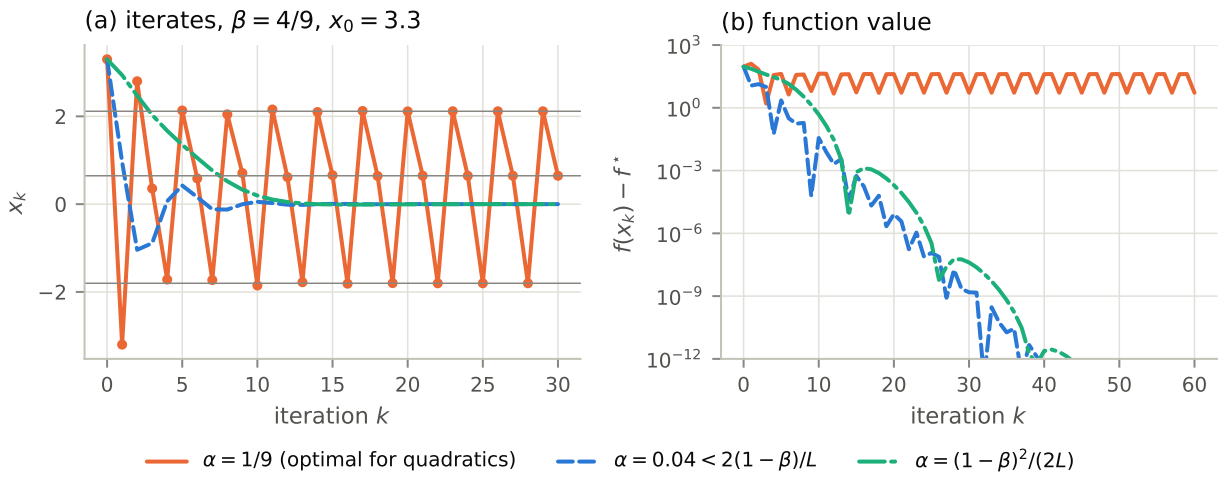


Figure 2: Heavy ball on the strongly convex function of Example 7.5, $\beta = \frac{4}{9}$, $x_{-1} = x_0 = 3.3$, exact gradients. (a) With the quadratic-optimal $\alpha = \frac{1}{9}$ the iterates are attracted to the 3-cycle $\{p_1, p_2, p_3\}$ (grey lines). (b) The function value then oscillates forever, while the step sizes allowed by Proposition 7.3 ($\alpha = 0.04$) and by Theorem 5.3 ($\alpha = (1-\beta)^2/(2L)$) converge.

So $p_1 \to p_2 \to p_3 \to p_1 \to \cdots$ is a periodic orbit: from $x_{-1} = p_3$, $x_0 = p_1$ the iterates cycle forever and $f(x_k) \geq f(p_1) \approx 5.2$ for all k, although f is strongly convex and smooth. Moreover the cycle attracts nearby trajectories: from $x_{-1} = x_0 = 3.3$ the iterates converge to it (Figure 2; [8] reports this for all $x_0 = x_{-1} \in [3.07, 3.46]$).

The parameters of Example 7.5 lie inside the quadratic stability region but violate all of our conditions: $\alpha = \frac{1}{9}$ exceeds $\frac{2(1-\beta)}{L} = \frac{2}{45}$ (Proposition 7.3), $\frac{1-\beta}{2L} = \frac{1}{90}$ (Theorem 6.8) and $\frac{(1-\beta)^2}{2L} = \frac{1}{162}$ (Theorem 5.3). With $\beta = \frac{4}{9}$ and $\alpha = 0.04 < \frac{2}{45}$ or $\alpha = \frac{1}{162}$, the same function and starting point give fast convergence (Figure 2).

7.5 The map of the parameter plane

Figure 3 collects the step-size conditions as regions in the (β, α) plane, nested from the most special setting to the most general one:

R4 quadratic f, exact gradients: convergence if and only if $\alpha L < 2(1+\beta)$ (Proposition 7.2);

R3 any L-smooth f, exact gradients: $\alpha L < 2(1-\beta)$ suffices (Proposition 7.3);

R2 convex f, stochastic gradients: $\alpha L \leq \frac{1-\beta}{2}$ suffices (Theorem 6.8);

R1 nonconvex f, stochastic gradients: $\alpha L \leq \frac{(1-\beta)^2}{2}$ suffices (Theorems 5.3 and 6.1).

Panel (b) shows the same regions in terms of the effective step size. There the picture is strikingly simple: R3 is $\gamma L < 2$ and R2 is $\gamma L \leq \frac{1}{2}$, both independent of β—momentum with effective step γ is as safe as gradient descent with step γ—whereas R1 shrinks like $\gamma L \leq \frac{1-\beta}{2}$, and R4 (quadratics) grows without bound as $\beta \to 1$. The star (Example 7.5) lies in R4 but outside R3, and the method indeed fails there.

We stress that R1–R3 are *sufficient* conditions. In particular we do not claim that R1 is necessary for stochastic nonconvex problems; it is the price of a short, general proof, and in practice larger

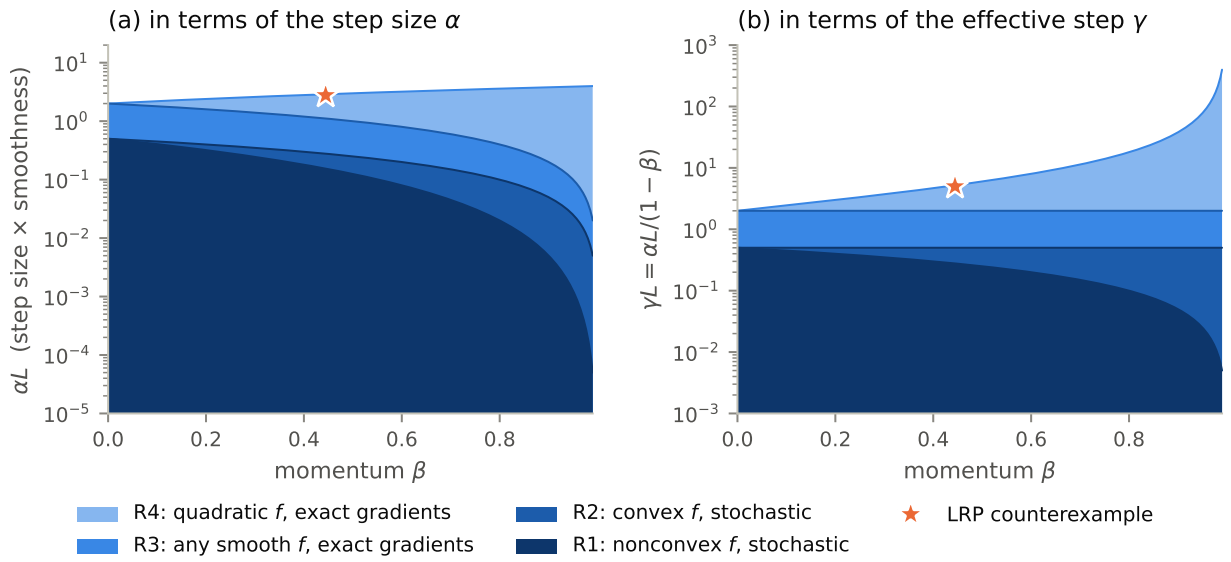


Figure 3: Where each guarantee holds. Nested regions of admissible parameters, (a) in terms of αL, (b) in terms of the effective step $\gamma L = \alpha L/(1-\beta)$ (log scales). R1: Theorem 5.3 (nonconvex, stochastic); R2: Theorem 6.8 (convex, stochastic); R3: Proposition 7.3 (any smooth f, exact gradients); R4: Proposition 7.2 (quadratics, exact gradients). The star marks the parameters of Example 7.5, for which heavy ball cycles on a strongly convex function.

steps are often used successfully. What the figure does show is that the question "does SGDM converge?" has different answers for different function classes, and that conclusions drawn from quadratics (R4) do not transfer to general functions.

7.6 Does momentum help?

The results above give a nuanced answer.

- *At equal effective step size, SGDM behaves like SGD.* The bounds of Section 6 coincide with the SGD bounds up to constants when α is replaced by γ; the noise floor is the same to first order (Proposition 7.1); and in experiments the trajectories are nearly indistinguishable (Figure 4(a)). This matches the conclusion of [9] that SGDM converges as fast as SGD.
- *At equal raw step size, momentum is a larger step.* It multiplies the effective step by $1/(1-\beta)$: faster initial progress, but a floor that is $1/(1-\beta)$ times higher (Figure 4(b)). Increasing β without decreasing α is therefore not "free".
- *No worst-case acceleration.* In our analysis the admissible effective step shrinks as $\beta \to 1$ (R1), and the PL rate $1 - \frac{\gamma\mu}{4}$ is no better than that of SGD. Acceleration (Section 7, "Momentum on quadratics") relies on quadratic structure and exact gradients; the quadratic-optimal tuning can fail on a smooth strongly convex function (Example 7.5) and, for stochastic gradients, there are problems on which the stochastic heavy-ball method provably does not improve on SGD [7].

The practical popularity of momentum is therefore better explained by its behaviour on problems that are locally close to quadratic and ill-conditioned, where the gradient noise is small relative to the signal, than by worst-case guarantees.

7.7 When the assumptions fail

- $\beta \geq 1$ *(no friction).* The roots of p in Proposition 7.2 have product $\beta \geq 1$, so at least one has modulus ≥ 1: heavy ball does not converge even on $\frac{1}{2}x^2$. At $\beta = 1$ the extrapolated point and the effective step size are not even defined. The condition $\beta < 1$ is necessary.
- *Noise growing with the gradient*, $\mathbb{E}_k\|g_k - \nabla f(x_k)\|^2 \leq \sigma^2 + M\|\nabla f(x_k)\|^2$: all results remain valid with the step-size condition tightened by the factor $1+M$. For the results based on Theorem 5.3 see Remark 5.8(ii); in the convex Theorem 6.8, the bound $\mathbb{E}_k\|g_k\|^2 \leq 2L(1+M)(f(x_k)-f^\star)+\sigma^2$ replaces $2\gamma(1-L\gamma)$ by $2\gamma(1-L\gamma(1+M))$, so $L\gamma(1+M) \leq \frac{1}{2}$ suffices.
- *Infinite variance (heavy-tailed noise).* Assumption A4 fails and the bounds above are vacuous; techniques such as gradient clipping are then used. This is beyond the scope of this report.
- *Biased gradients.* Unbiasedness (Assumption A3) is used in Lemmas 5.1 and 5.2. With a bias, both lemmas acquire additional error terms and convergence can only be guaranteed up to a neighbourhood whose size depends on the bias.
- *Non-smooth f.* Without Assumption A1 there is no descent lemma; momentum methods for non-smooth (weakly convex) problems require different tools, see [10].

7.8 Summary: a checklist

SGDM achieves the SGD-style expected decrease, and hence the convergence guarantees of Section 6, under the following circumstances.

1. **The problem:** f is L-smooth and bounded below; the stochastic gradients are unbiased with bounded variance σ^2 (Assumptions A1 to A4).
2. **The momentum:** $\beta \in [0, 1)$—friction must be present.
3. **The step size, measured as the effective step $\gamma = \alpha/(1-\beta)$:**
   - general nonconvex, stochastic: $\gamma \leq \frac{1-\beta}{2L}$, i.e. $\alpha \leq \frac{(1-\beta)^2}{2L}$;
   - convex, stochastic: $\gamma \leq \frac{1}{2L}$;
   - any smooth f, exact gradients: $\gamma < \frac{2}{L}$.
4. **The target:** with a constant step one reaches a noise region of size $O(\gamma\sigma^2)$, which is unavoidable (Proposition 7.1); to reach accuracy ε^2 in $\min_k \mathbb{E}\|\nabla f(x_k)\|^2$ choose $\gamma \propto 1/\sqrt{K}$ (Corollary 6.2), or use decaying steps (Theorem 6.4) for asymptotic convergence.
5. **Linear rates** need additional structure such as the PL inequality or strong convexity (Theorem 6.6); the rate is $1 - \gamma\mu/4$.
6. **Beware of quadratic intuition:** parameters that are stable or even optimal for quadratics can make the method cycle on a strongly convex function (Example 7.5).

8 Numerical illustrations

The experiments below illustrate the theorems; they are not needed for the proofs. All of them run Algorithm 1 in heavy-ball form (the implementation in `experiments/common.py` is a direct transcription of the update $v_{k+1} = \beta v_k - \alpha_k g_k$, $x_{k+1} = x_k + v_{k+1}$), vectorized over independent

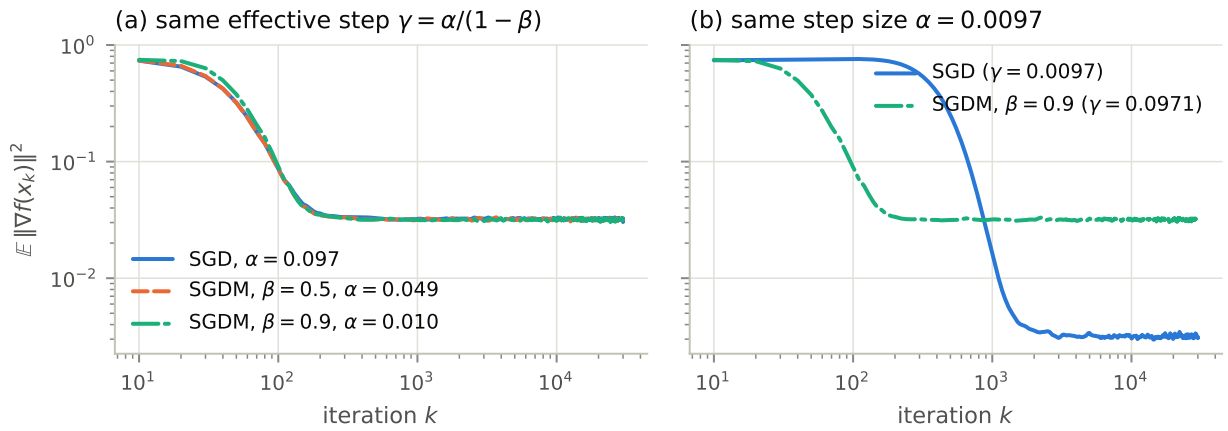


Figure 4: SGD versus SGDM on the nonconvex logistic regression problem (P3), mean of $\|\nabla f(x_k)\|^2$ over 100 runs (lightly smoothed). (a) Same effective step $\gamma = \frac{1-0.9}{2L}$, inside the range of Theorem 5.3 for every β shown: the three methods are indistinguishable. (b) Same raw step $\alpha = \frac{(1-0.9)^2}{2L}$: momentum $\beta = 0.9$ makes the effective step 10 times larger, so it converges faster at first but stalls at a 10 times higher noise floor.

runs; expectations are estimated by averaging over runs. Every figure is produced by one script (Appendix B).

Test problems.

(P1) *Quadratic:* $f(x) = \frac{h}{2}x^2$ with $h = L = 1$; stochastic gradient $g = hx + \xi$, $\xi \sim \mathcal{N}(0, \sigma^2)$.

(P2) *Nonconvex PL function:* $f(x) = \sum_{i=1}^{10}(x_i^2 + 3\sin^2 x_i)$ on $\mathbb{R}^{10}$ ($L = 8$, $\mu = \frac{1}{32}$, $f^\star = 0$, Lemma 2.7(c)); stochastic gradient $g = \nabla f(x) + \xi$ with $\xi \sim \mathcal{N}(0, \frac{\sigma^2}{10} I)$, so that $\mathbb{E}\|\xi\|^2 = \sigma^2$.

(P3) *Nonconvex logistic regression:* $f(x) = \frac{1}{n}\sum_{i=1}^{n} \log(1 + e^{-b_i a_i^\top x}) + \lambda \sum_{j=1}^{d} \frac{x_j^2}{1+x_j^2}$ with $n = 1000$, $d = 20$, $\lambda = 0.1$, features $a_i \sim \mathcal{N}(0, I_d)$, and labels $b_i \in \{\pm 1\}$ drawn from the logistic model with a random parameter vector. The regularizer is nonconvex. Stochastic gradients use one uniformly sampled example (mini-batch size 1); they are unbiased with bounded variance, and f is L-smooth with $L \le \frac{\|A\|_2^2}{4n} + 2\lambda \approx 0.515$, where A has rows $a_i^\top$. All runs start at $x_0 = (-1, \dots, -1)$.

(P4) *The function of Example 7.5* (strongly convex, exact gradients).

Potential versus function value (Figure 1). On (P1) with $\beta = 0.9$ and the largest step allowed by Theorem 5.3, $\alpha = (1-\beta)^2/(2L) = 0.005$, the function value $f(x_k)$ is not monotone even without noise, because the iterate overshoots (panel (b)); the potential Φ_k decreases at every step, as guaranteed by Corollary 5.6. (The script also checks this monotonicity numerically.) With noise (panel (c)), the empirical floor of $\mathbb{E}f(x_k)$ is 0.01252, matching the exact value of Proposition 7.1 to four digits.

SGD versus SGDM (Figure 4). On (P3), SGD and SGDM with $\beta \in \{0.5, 0.9\}$ run at the same effective step size produce practically identical curves (panel (a)), including the level of the noise floor: SGDM behaves like SGD with step γ, as the parallel between Lemma 3.1 and Theorem 5.3

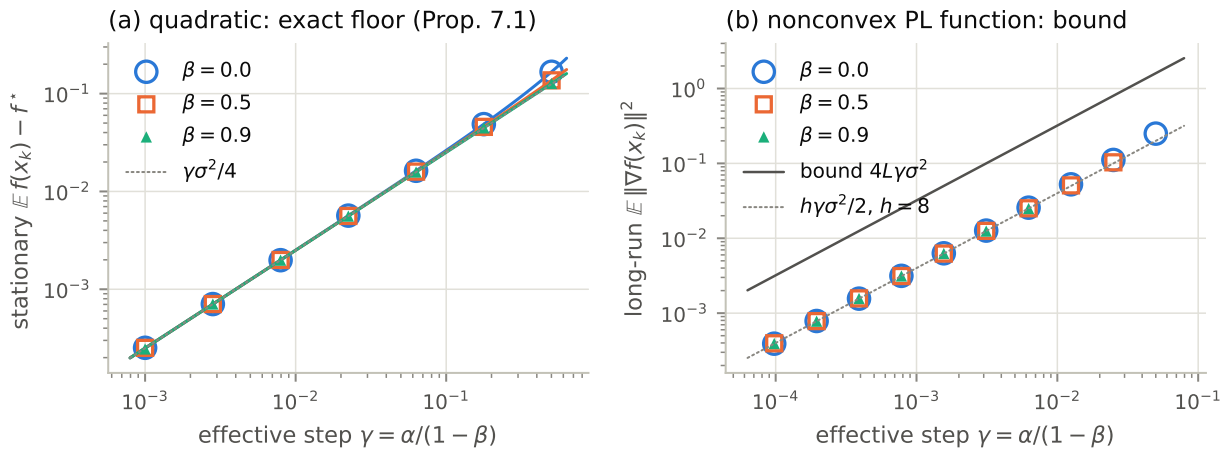


Figure 5: The noise floor with a constant step scales like $\gamma\sigma^2$ and, to first order, depends on (α, β) only through γ (markers for different β lie on top of each other). (a) Quadratic (P1), $\sigma = 1$: empirical stationary $\mathbb{E}f(x_k)$ (markers; 4000 runs) and the exact formula (7.1) (lines). (b) Nonconvex PL function (P2), $\sigma = 1$: long-run $\mathbb{E}\|\nabla f(x_k)\|^2$ (400 runs) for step sizes inside the range of Theorem 5.3, the bound $4L\gamma\sigma^2$ of Theorem 6.1 (as $K \to \infty$), and the local quadratic prediction $h\gamma\sigma^2/2$ with $h = f''(0) = 8$.

suggests. At the same raw step size (panel (b)), momentum is simply a ten times larger step: faster initial progress, ten times higher floor, consistent with Proposition 7.1.

The noise floor (Figure 5). For each $\beta \in \{0, 0.5, 0.9\}$ and a range of effective step sizes we run the method from the minimizer until it reaches stationarity and average the quantity of interest over time and runs. On the quadratic, the measurements agree with (7.1) to within 2% for every (γ, β). On the nonconvex PL function, the measured floors lie on the line $h\gamma\sigma^2/2$ predicted by the local quadratic approximation at the minimizer, and about a factor 8 below the bound $4L\gamma\sigma^2$ of Theorem 6.1: the theorem has the correct dependence on γ and σ, with a pessimistic constant. In both panels the points for different β coincide.

Diminishing steps (Figure 6). With a constant step the error stalls at its noise floor; with the schedules of Theorem 6.4 and Corollary 6.7 it keeps decreasing. On the logistic problem the squared gradient norm decreases roughly like $k^{-1/2}$, which is the order of γ_k; on the PL function the optimality gap eventually decreases like $1/k$, as predicted by Corollary 6.7. (The theoretical schedule for the PL function starts slowly because the certified PL constant $\mu = \frac{1}{32}$ is much smaller than the local curvature at the minimizer.) Note also the non-monotone transient around $k \approx 200$ in panel (b), another manifestation of overshooting.

The counterexample (Figure 2). On the function of Example 7.5, quadratic-optimal parameters lead to the predicted 3-cycle, while the step sizes allowed by Proposition 7.3 and Theorem 5.3 converge. The script verifies the cycle in exact rational arithmetic.

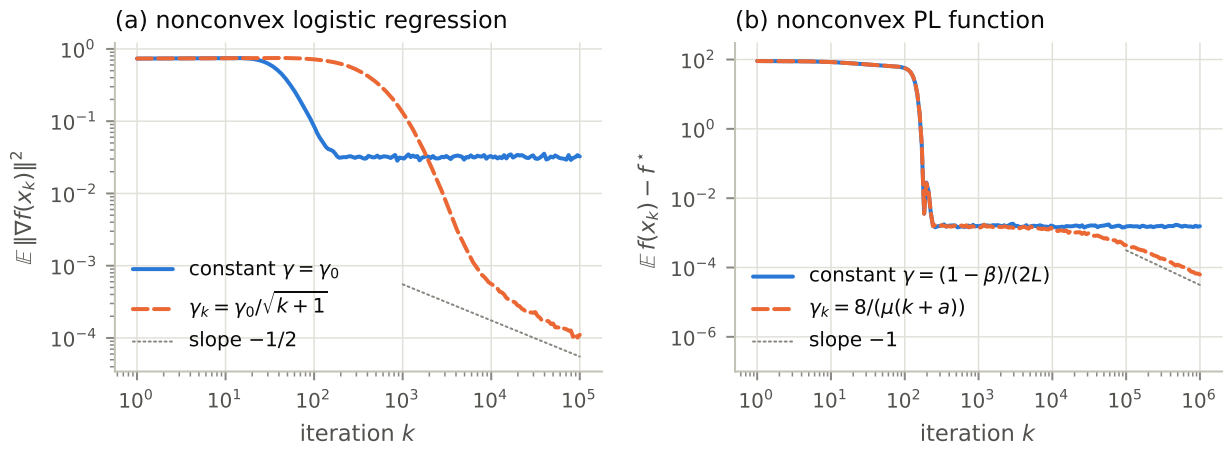


Figure 6: Diminishing step sizes remove the noise floor ($\beta = 0.9$; both schedules start from the same $\gamma_0 = \frac{1-\beta}{2L}$). (a) Nonconvex logistic regression (P3), 50 runs: constant step versus $\gamma_k = \gamma_0/\sqrt{k+1}$ (Theorem 6.4(c)). (b) Nonconvex PL function (P2), 100 runs: constant step versus $\gamma_k = \frac{8}{\mu(k+a)}$ with $a = \frac{16L}{\mu(1-\beta)}$ (Corollary 6.7); the dotted reference lines have slope $-\frac{1}{2}$ in (a) and -1 in (b).

9 Conclusion

The standard analysis of SGD rests on a one-step expected decrease of the function value. For SGD with momentum the function value is the wrong yardstick, because momentum makes the iterates overshoot. The report's main message fits in three sentences.

1. *SGDM is SGD on the extrapolated points.* The points $z_k = x_k + \frac{\beta}{1-\beta}(x_k - x_{k-1})$, where the ball would come to rest if the force were switched off, follow exact SGD steps $z_{k+1} = z_k - \gamma g_k$ with the effective step size $\gamma = \alpha/(1-\beta)$; only the gradient is evaluated at the nearby point x_k (Lemma 4.5).
2. *A potential decreases in expectation, exactly as f does for SGD.* The potential $\Phi_k = f(z_k) - f^\star + \frac{L\beta^2}{2(1-\beta)^2}\|x_k - x_{k-1}\|^2$ satisfies $\mathbb{E}_k\Phi_{k+1} \le \Phi_k - \frac{\gamma}{4}\|\nabla f(x_k)\|^2 + L\gamma^2\sigma^2$ whenever $\alpha \le (1-\beta)^2/(2L)$: friction must dissipate kinetic energy faster than the gradient mismatch creates error (Theorem 5.3).
3. *Everything else follows as for SGD.* Telescoping gives the $O(1/\sqrt{K})$ nonconvex rate, diminishing steps give asymptotic convergence, the PL inequality gives linear convergence to a noise floor, and convexity relaxes the step-size condition to $\gamma \le 1/(2L)$ (Section 6).

As for *when* convergence holds (Section 7): the momentum must satisfy $\beta < 1$; the step size must be small in terms of the effective step γ, with a threshold that depends on the function class (Figure 3); a constant step only reaches a noise floor of order $\gamma\sigma^2$, which is unavoidable and the same as for SGD (Proposition 7.1); and stability or optimality on quadratics does not imply convergence on general functions (Example 7.5). Once step sizes are compared on the right scale, our upper bounds for SGDM match those for SGD in the dominant noise term, up to constants, and they show no acceleration.

A Linear recurrences and powers of 2×2 matrices

These two standard facts are used in Propositions 7.1 and 7.2.

Lemma A.1 (Second-order linear recurrences)**.** *Let $a \in \mathbb{R}$, $b \neq 0$, and let $\lambda_1, \lambda_2 \in \mathbb{C}$ be the roots of $p(\lambda) = \lambda^2 - a\lambda + b$. The solutions $(x_k)_{k \geq -1}$ of $x_{k+1} = ax_k - bx_{k-1}$ ($k \geq 0$) are exactly the sequences*

(i) *$x_k = c_1\lambda_1^k + c_2\lambda_2^k$ with $c_1, c_2 \in \mathbb{C}$, if $\lambda_1 \neq \lambda_2$;*

(ii) *$x_k = (c_1 + c_2 k)\lambda_1^k$ with $c_1, c_2 \in \mathbb{C}$, if $\lambda_1 = \lambda_2$.*

Proof. These are solutions. For a root λ, $\lambda^{k+1} - a\lambda^k + b\lambda^{k-1} = \lambda^{k-1}p(\lambda) = 0$, and sums of solutions are solutions. In case (ii), $\lambda_1 = a/2$ and $b = \lambda_1^2$ (product of the roots), and

$$(k+1)\lambda_1^{k+1} - ak\lambda_1^k + b(k-1)\lambda_1^{k-1} = \lambda_1^{k-1}\big[k\,p(\lambda_1) + \lambda_1^2 - b\big] = 0.$$

There are no others. A solution is determined by (x_{-1}, x_0), so it suffices to show that every pair (x_{-1}, x_0) is attained. In case (i) we need $c_1\lambda_1^{-1} + c_2\lambda_2^{-1} = x_{-1}$, $c_1 + c_2 = x_0$; the determinant of this linear system is $\lambda_2^{-1} - \lambda_1^{-1} = \frac{\lambda_1 - \lambda_2}{\lambda_1\lambda_2} \neq 0$ (the roots are nonzero because $\lambda_1\lambda_2 = b \neq 0$). In case (ii) we need $(c_1 - c_2)\lambda_1^{-1} = x_{-1}$ and $c_1 = x_0$, which has the solution $c_1 = x_0$, $c_2 = x_0 - \lambda_1 x_{-1}$. □

When $b = \beta = 0$ the heavy-ball recurrence reduces to $x_{k+1} = ax_k$, with solutions $x_k = a^k x_0$; the arguments of Proposition 7.2 apply verbatim with the roots a and 0.

Lemma A.2 (Powers of a 2×2 matrix)**.** *Let M be a real 2×2 matrix whose eigenvalues satisfy $\max\{|\lambda_1|, |\lambda_2|\} < \rho < 1$. Then there is a constant C_M such that every entry of M^k is at most $C_M\rho^k$ in absolute value, for all $k \geq 0$. Consequently, for any 2×2 matrices C_0, Q, we have $M^k C_0 (M^\top)^k \to 0$, and the series $\sum_{j \geq 0} M^j Q (M^\top)^j$ converges.*

Proof. If $\lambda_1 \neq \lambda_2$, M is diagonalizable over $\mathbb{C}$: $M = V\,\mathrm{diag}(\lambda_1, \lambda_2)V^{-1}$, so $M^k = V\,\mathrm{diag}(\lambda_1^k, \lambda_2^k)V^{-1}$ and its entries are bounded by a constant times $\max_i |\lambda_i|^k \leq \rho^k$. If $\lambda_1 = \lambda_2 = \lambda$, then either $M = \lambda I$, or $M = VJV^{-1}$ with the Jordan block $J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$, for which $J^k = \begin{pmatrix} \lambda^k & k\lambda^{k-1} \\ 0 & \lambda^k \end{pmatrix}$; since $k|\lambda|^{k-1}/\rho^k \to 0$, the entries of M^k are again bounded by a constant times ρ^k. The entries of $M^j Q (M^\top)^j$ are then bounded by a constant times ρ^{2j}, so the series converges absolutely, and similarly $M^k C_0 (M^\top)^k \to 0$. □

B Reproducing the figures

The repository contains the LaTeX source of this report (`report/`) and the Python code for all figures (`experiments/`, requiring `numpy` and `matplotlib`). From the repository root:

```
pip install -r experiments/requirements.txt
python experiments/run_all.py          # regenerates report/figures/*.pdf
cd report && latexmk -pdf main.tex     # rebuilds the PDF
```

or use the `Makefile`: `make figures` reruns the experiments and `make` rebuilds the PDF and copies it to `SGDM_Convergence_Report.pdf`. The scripts and the figures they produce are:

script	figure	content
`potential.py`	Figure 1	$f(x_k)$ versus Φ_k; checks monotonicity of Φ_k
`regions.py`	Figure 3	parameter regions
`sgd_vs_sgdm.py`	Figure 4	SGD versus SGDM on (P3)
`noise_floor.py`	Figure 5	noise floor versus γ
`diminishing.py`	Figure 6	diminishing step sizes
`counterexample.py`	Figure 2	the 3-cycle of Example 7.5, verified exactly

All random seeds are fixed, so the figures are reproducible.

References

[1] D. P. Bertsekas and J. N. Tsitsiklis. Gradient convergence in gradient methods with errors. *SIAM Journal on Optimization*, 10(3):627–642, 2000.

[2] L. Bottou, F. E. Curtis, and J. Nocedal. Optimization methods for large-scale machine learning. *SIAM Review*, 60(2):223–311, 2018.

[3] S. Gadat, F. Panloup, and S. Saadane. Stochastic heavy ball. *Electronic Journal of Statistics*, 12(1):461–529, 2018.

[4] E. Ghadimi, H. R. Feyzmahdavian, and M. Johansson. Global convergence of the heavy-ball method for convex optimization. In *Proceedings of the European Control Conference (ECC)*, 2015.

[5] S. Ghadimi and G. Lan. Stochastic first- and zeroth-order methods for nonconvex stochastic programming. *SIAM Journal on Optimization*, 23(4):2341–2368, 2013.

[6] H. Karimi, J. Nutini, and M. Schmidt. Linear convergence of gradient and proximal-gradient methods under the Polyak–Łojasiewicz condition. In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD)*, 2016.

[7] R. Kidambi, P. Netrapalli, P. Jain, and S. M. Kakade. On the insufficiency of existing momentum schemes for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2018.

[8] L. Lessard, B. Recht, and A. Packard. Analysis and design of optimization algorithms via integral quadratic constraints. *SIAM Journal on Optimization*, 26(1):57–95, 2016.

[9] Y. Liu, Y. Gao, and W. Yin. An improved analysis of stochastic gradient descent with momentum. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[10] V. V. Mai and M. Johansson. Convergence of a stochastic gradient method with momentum for non-smooth non-convex optimization. In *International Conference on Machine Learning (ICML)*, 2020.

[11] B. T. Polyak. Some methods of speeding up the convergence of iteration methods. *USSR Computational Mathematics and Mathematical Physics*, 4(5):1–17, 1964.

[12] H. Robbins and S. Monro. A stochastic approximation method. *The Annals of Mathematical Statistics*, 22(3):400–407, 1951.

[13] H. Robbins and D. Siegmund. A convergence theorem for non negative almost supermartingales and some applications. In *Optimizing Methods in Statistics*, pages 233–257. Academic Press, 1971.

[14] O. Sebbouh, R. M. Gower, and A. Defazio. Almost sure convergence rates for stochastic gradient descent and stochastic heavy ball. In *Conference on Learning Theory (COLT)*, 2021.

[15] I. Sutskever, J. Martens, G. Dahl, and G. Hinton. On the importance of initialization and momentum in deep learning. In *International Conference on Machine Learning (ICML)*, 2013.

[16] T. Yang, Q. Lin, and Z. Li. Unified convergence analysis of stochastic momentum methods for convex and non-convex optimization. arXiv:1604.03257, 2016.

[17] H. Yu, R. Jin, and S. Yang. On the linear speedup analysis of communication efficient momentum SGD for distributed non-convex optimization. In *International Conference on Machine Learning (ICML)*, 2019.

[18] S. K. Zavriev and F. V. Kostyuk. Heavy-ball method in nonconvex optimization problems. *Computational Mathematics and Modeling*, 4:336–341, 1993.